Course Book

INTRODUCTION TO ACADEMIC WORK

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INTERNATIONAL
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INTRODUCTION TO ACADEMIC WORK

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INTRODUCTION

WELCOME

SIGNPOSTS THROUGHOUT THE COURSE BOOK

This course book contains the core content for this course. Additional learning materials can be found on the learning platform, but this course book should form the basis for your learning.

The content of this course book is divided into units, which are divided further into sections. Each section contains only one new key concept to allow you to quickly and efficiently add new learning material to your existing knowledge.

At the end of each section of the digital course book, you will find self-check questions. These questions are designed to help you check whether you have understood the concepts in each section.

For all modules with a final exam, you must complete the knowledge tests on the learning platform. You will pass the knowledge test for each unit when you answer at least 80% of the questions correctly.

When you have passed the knowledge tests for all the units, the course is considered finished and you will be able to register for the final assessment. Please ensure that you complete the evaluation prior to registering for the assessment.

Good luck!

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LEARNING OBJECTIVES

Introduction to Academic Work explains the basics of scientific theory and presents the most important aspects of good scientific practice. The building blocks of fundamental academic knowledge include an introduction to research methods and mechanisms.

This introduction into academic work provides an overview of the most important components of academic writing that the student can practice in real-world lessons. These lessons then introduce the different types of IU International University of Applied Sciences exams, providing insight into their requirements and implementation.

This combination of theoretical principles and practical execution lays the foundation for the future of scientific work.

UNIT 1

THEORY OF SCIENCE

STUDY GOALS

On the completion of this unit, you will have learned ...

- the essential characteristics of scientific research.
- how to distinguish between different fundamental research assumptions.
- how to identify core research decisions.

1. THEORY OF SCIENCE

Case Study

Simon is studying business administration at IU International University of Applied Sciences (IU) while also working at zielNET, a small market research company. Four years ago, he successfully completed his training as a marketing and social research specialist at zielNET. During that time, he completed a written assignment that focused on how different customer groups formed their opinions.

Simon loves getting to the bottom of things, as Goethe's Faust (1808/2005) says, "To know what holds the world together at its core". Having a particular interest in analyzing target groups and group behavior is what led him to his current profession. Simon's supervisor appreciates this analytical quality and often turns to him for advice on how to approach specific consumer groups.

Currently, Simon is involved in an important project for zielNET: analyzing the recent product failure of a long-time major customer. Despite the fact that zielNET invested six months in the product's relaunch, customers still appear to have no interest in purchasing it. Simon now tries to address this issue empirically in his current course. What does it actually mean to work scientifically? He is trying to remember models and theories that dealt with consumer decisions and buying behavior of certain target groups and that could be helpful for investigating the issue at hand.

1.1 Introduction to Science and Research

This section begins with some key reflections on research and addresses scientific theory. This may seem tedious and complicated at first, but it should quickly become clear that the underlying structures of science have a major influence on our everyday lives. Additionally, analytical tools will be provided to enable a critical examination of results of future research projects and their fundamental assumptions.

Case Study: Observation, Reflection, and Reasoning

The following situation serves as a starting point for further discussion. Thousands of years ago, a hunter and gatherer was out collecting branches of wood. On the way back to his village, he stumbled and the branches fell to the ground. Understandably, his initial reaction was one of annoyance. However, he began to compare this situation with other previous experiences and realized that things actually fall down again and again, but nothing ever falls up. He continued to reflect and realized that this applies to all the things around him: tree branches, animals, stones, fruit, etc. Even the leaves that he recently collected for the shelter floor had fallen down, albeit more slowly. Some leaves were even blown away by the wind, an event which he had never observed with stones. At this point he realized that everything he collects can fall down—even if the speed of doing so differs—but nothing has the ability to fall up. He now wonders if other members of his tribe are aware of this, or if they know of anything that falls up.

Without being able to provide answers to these questions, one could proceed as follows: First, the tribal elder could be questioned along with some other tribal members about their experiences with falling objects. However, their opinions may diverge. For example, the tribal elder could remind him that the souls of the deceased ascend to the gods, i.e., fall up, not down. Other tribal members might claim to have seen leaves fly up due to a particularly strong wind. While there might be a few who swear that there are objects that do fall upward, on the whole, there is agreement that the tendency is for most things to fall down, with estimates of speed varying greatly.

Although these assessments point us in the right direction, many questions remain unanswered. It is now necessary to consider how the whole phenomenon could be subjected to a more detailed investigation.

The Science Council (2009) defines science as “the pursuit and application of knowledge and understanding of the natural and social world following a systematic methodology based on evidence”. Science can also be defined as, “[t]he intellectual and practical activity encompassing the systemic study of the structure and behavior of the physical and natural world through observation and experiment” (Lexico.com, 2019a).

Science usually means the process of research and, more precisely, scientific research. The following definition applies: “Scientific research is conducted within the rules and conventions of science” (Veal, 2018, p. 6).

This means that scientific research is a systematic, rule-based process used to gain knowledge. In the case above, the initial observation (branches fall down) cannot be considered to be at the level of scientific research yet; however, following a systematic and established process brings us closer to this goal.

Based on scientific concepts broadly accepted today, such as the fact that gravity exists, one can find out a lot about the subject of gravity. In fact, due to the ubiquitous availability of information, some of these explanatory approaches and theories can now be regarded as established knowledge. Therefore, one would certainly agree that the explanation of gravity can be regarded as true. But is that explanation incontrovertible? At this point, one can refer again to the tribal elder talking about the ascending souls of the deceased. Can people be absolutely sure that such a thing does not exist? The question of whether something is considered true depends on the basic assumptions of our worldview.

It is known, for example, that witness statements after accidents or crimes are often unreliable because different witnesses can have completely different recall about the event itself. One person swears that he heard three gun shots, while another person at the scene says she heard only one gun shot. The vehicle fleeing the scene that caused the accident was a blue SUV, says one person, while to someone else it appeared to be a black station wagon. Each of these individuals is absolutely convinced that they are telling the truth.

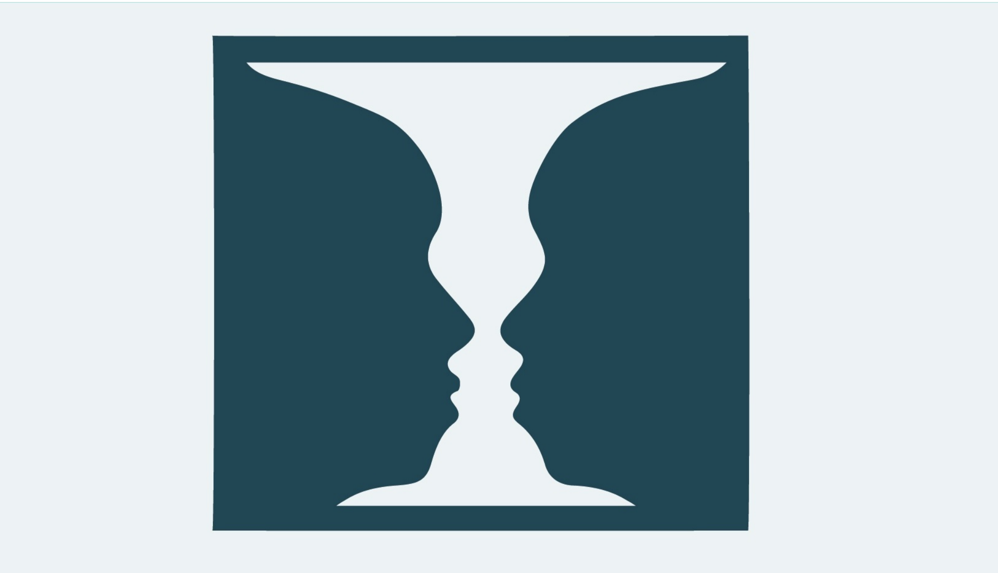
Indeed, there are many situations that people accept as true and at the same time remain questionable, depending on certain basic assumptions people share. Here are a few examples.

Example: True blue

When someone refers to a blue house, there is an unspoken assumption that the house they are referring to is painted blue. However, for a person who is color-blind, this description is not their truth. Here it is assumed that the color-blind person is wrong, because they cannot recognize the color blue. But is this really the case? Do people think of this person as ill because they cannot see color? What would happen if color-blind people were actually the dominant group and everyone else was suffering from a "color vision" mutation? What if suddenly most people could no longer recognize colors and agreed that the few remaining people who could see them were themselves ill? Then the blue house would suddenly be something that does not even exist, since for the color-blind, it would be a shade of gray. The description of a blue house would no longer be true; rather, the truth—as generally shared knowledge—would be that the house is gray.

On the subject of "what is truth", the following cognitive optical illusion serves as an illustration:

Figure 1: Rubin's Vase



Source: Becker-Carus & Wendt, 2017, p. 124.

In the image of Rubin's vase, is it true that two faces are depicted or is it true that a vase is depicted? Or are both things true? This question of fact should be differentiated from the question of what the observer sees. On closer examination, the central scientific framework of observation also proves to be questionable.

Example: Global poverty

According to the United Nations Department of Economic and Social Affairs (2019), extreme poverty has risen in the United States; in Europe there has been an overall decline, but the number of poor people is still higher than prior to the 2008 financial crisis. In many countries, the gap between rich and poor continues to widen. For this purposes, a “relative concept” of poverty is used when discussing this topic. Relative poverty is “determined by income distribution over a given population and defined according to societal norms” (Open Education Sociology Dictionary, 2020). According to the relative concept of poverty, someone is regarded as poor if that person earns less than 50 percent of the median income in the society they live in. This seems plausible and, according to the specific definition given, is one true way of measuring poverty.

At this point, it should neither be discussed whether this concept of poverty makes sense, nor it should be denied that poverty is lamented in our societies. What is important here is that this particular definition of poverty creates truths that are politically relevant and can also be questioned. If in a country with 15 percent poverty, for example, all salaries were doubled from one day to the next, the country would still have a deplorable poverty level of 15 percent. Mathematically this is correct, but is it true? This up for discussion.

The examples of color perception and poverty are intended to raise awareness to the fact that scientific research produces results that are generally regarded as true. However, this truth is always based on certain assumptions that must be considered when trying to understand and classify the results achieved. Statistics play a major role in societal and political contexts; however, it is often very difficult to evaluate the validity of “hard facts” because information is lacking.

Given all these complex requirements, it seems quite challenging to become a successfull scholar; especially, when thinking of big names such as Einstein, Galileo Galilei, Marie Curie, and Maria Goeppert Mayer. However, research often means going “two steps forward, one step back”—a slow and steady process that allows even novice researchers to be successful.

Scientific research produces findings that can be regarded as true in the sense that researchers’ assumptions are known and shared. Nutritional recommendations over the last 60 years, for example, have been based on scientific research. Meat diets, carbohydrate diets, low-carb programs, interval fasting, therapeutic fasting, moderate alcohol consumption, no alcohol consumption, the Mayo diet, etc., all came about because of scientific recommendations and findings that resulted in facts being recognized as “truth”.

However, scientific findings are also opportunities for critique. Criticism is an extremely important part of the scientific research process because it gives rise to new ideas for future research projects. During this course you will acquire the tools to understand research, critically question it, and develop your own research approaches.

With regard to the relevance of research, it can be stated that research findings are not only used to explain existing conditions, but also partly to predict future developments. Here are some examples of scientific questions:

- Why is Apple in a position to market mobile phones and computers at substantially higher prices than its competitors while offering essentially the same performance?
- How does international milk consumption affect theCO_2 balance of the environment and thus climate change?
- How can agile leadership in medium-sized production companies influence efficiency and productivity?

These examples convey a first impression of how complex open questions can be, including all their conceivable sub-aspects, in the broad field of science and research.

1.2 Research Paradigms

Research, and the way in which scientific research is approached, has a lot to do with underlying assumptions. There is a need to review the steps that prior studies have taken and to also think about the basic assumptions of the own research project. These basic assumptions are also referred to in the literature as paradigms or, more precisely, as research paradigms. Research paradigms are the most fundamental convictions from which knowledge is gained in the research process (Guba & Lincoln, 1994; Saunders et al., 2019, pp. 138–151). Depending on the research question, different positions can be taken:

Scenarios A and B will be used for comparison:

A. How does the acceleration from 0 to 100 km/h of a vehicle change when special gasoline is used as fuel as compared to the use of normal gasoline?

B. How do leaders of social organizations evaluate their own ethical approach when determining an organization-wide ethical orientation, according to Swiss economist Peter Ulrich?

When having a look at both questions, it is immediately apparent that each question requires a different scientific approach to find the answer. In scenario A, an experiment will most likely be carried out in which the same vehicle will be refueled once with normal gasoline and once with special gasoline, the exact same acceleration conditions will be utilized, and a time measurement will be carried out. In addition, the testing conditions will be reproduced as exactly as possible to generate a reliable conclusion, i.e., weather conditions and subsurface conditions will be replicated.

In scenario B, attitudes of a particular group of people—leaders in social organizations—need to be investigated. The organizations themselves can vary greatly, as can the attitudes about ethics; the latter is based strongly on the values held by the survey respondents. The standardization of the research process tends to be more difficult than in scenario A as there are many influences on the respondents and on the survey situation that are difficult to standardize and control.

These two research scenarios illustrate different underlying scientific assumptions—research paradigms. In order to understand the path to knowledge for both scenarios, four questions are typically asked (Guba & Lincoln, 1994, pp. 107–108; Saunders et al., 2019, pp. 144–151):

- Ontology: What is truth? What can a person say is true?
- Epistemology: What can humans know?
- Methodology: What instruments can be used to gain knowledge?
- Influence of the researcher: To what extent does the researcher influence the research results?

When looking at these four questions, the differences between scenarios A and B become clear. For scenario A, it can be assumed that there must be an unambiguous result for the acceleration behavior of the different types of gas (ontology) and that this result is also exactly measurable and reproducible (epistemology). If the test arrangements are identical for both gas options, the results are not open for interpretation and can therefore be regarded as objectively true. Methodically, one will almost certainly conduct an experiment in which one leaves all conditions as similar as possible and only changes the gas used (methodology). In this respect, the influence of the researcher is kept out of the research process as much as possible (influence of the researcher).

In scenario B, while talking to leaders of organizations will definitely deliver relevant insights, the "truth" here will manifest itself in interpretation-related tendencies (ontology). Therefore, it will be difficult to arrive at a definitive and unequivocal conclusion (epistemology). In methodological terms, an experiment does not seem to apply to this scenario, rather one would most likely use open questions so that the managers can express themselves freely (methodology). In such a survey, influences of the researcher are hard to eliminate because factors such as the type of question, mood, preferences, and dislikes can be important influencing factors (influence of the researcher). The result is not a measurable value as in scenario A, but rather an assessment based on interpretation.

Scenario A is presumably based on an "explanatory research paradigm" (Saunders et al., 2019, p. 144–147; Gubrium, 2012, p. 417). In the literature, this paradigm is also called "positivism" and includes different versions. However, it is not necessary to go into more detail on this here. Scenario B is based on an "understanding research paradigm" (Guba & Lincoln, 1994, pp. 110–116; Gubrium, 2012, p. 417), often referred to as "constructivism", of which again there are different versions but will not be expanded upon here. The following table gives an overview of the most important points for each research paradigm.

Table 1: Fundamental Research Paradigms

Paradigm	Explanatory Paradigm	Understanding Paradigm
Ontology: What is truth? What can one say is true?	A person can make statements about reality (objects, living creatures, etc., in the world).	Reality occurs in humans through experiences, attitudes, and surroundings.

Paradigm	Explanatory Paradigm	Understanding Paradigm
Epistemology: What can humans know?	Objectifiable conditions: Insights and findings are measurable and also exist for multiple people.	Generally subjective insights and findings arise through interactions and interpretation processes.
Methodology: What instruments can be used to gain knowledge?	Experimental methods, standardized methods, quantitative methods	Interpretative methods, qualitative methods
Influence of the Researcher: To what extent does the researcher influence the research results?	Influence is excluded as much as possible.	Subjectivity of the researchers: They are involved in the research process because no interpretation would be possible without them.

Source: Created on behalf of IU (2020).

Thus, an explanatory research paradigm searches for universally valid laws by subjecting theories or assumptions to a test in order to either confirm (verify) or refute (falsify) their universality. Since reality here is regarded as non-individual—i.e., independent of the individual—one tries to exclude the subjectivity of the researcher as much as possible. In contrast, the understanding research paradigm assumes that reality is always constructed by the individual and arises through interpretation. In this respect, the subjectivity of the researcher is an integral part of the research process. An understanding research paradigm often motivates research fields in which little or no theoretical approaches exist.

Each research paradigm has a different impact on the further development of research strategies. It is important to note that no paradigm is considered “better” or “superior”, rather each research paradigm has its relevance in different fields of research, often even complementing each other.

1.3 Research Decisions

Research paradigms determine the way research is conducted. Furthermore, research questions often already allow to make conclusions about the underlying assumptions. For a research question to be asked, the central ideas for carrying out the investigation must be formulated in advance. There are several research decisions to be considered which is also referred to as research design. The following decisions, among others, have to be made (Bazeley, 2004, p. 141; Bell & Waters, 2018, pp. 23–46; Creswell & Creswell, 2018, pp. 3–21):

- type of research strategy
- type of scientific reasoning
- type of data
- type of research to be carried out

Type of Research Strategy

This research decision is aimed at the question and choice of methodology, of whether research should be quantitative or qualitative. Research is primarily about collecting, analyzing, and interpreting data, data that sometimes can be very different in nature. **Quantitative research** aims at results that can be expressed in data, e.g., sales figures or average length of stay in health care facilities (Saunders et al., 2019, pp. 176–178). In quantitative research, a mathematically comprehensible, mathematically "correct" result is obtained and thus the research arrives at a statement whose accuracy — assuming mathematically precise procedures were followed — cannot be refuted. However, the interpretative aspect of quantitative research is often "hidden" in the translation or reduction of complex terms into numerical scales (e.g., the term "quality of life" in clinical studies on patients at the end of their lives). In quantitative research, representativeness is also important, i.e., the extent to which the result obtained for a sample can be generalized to the population as a whole. A good example is election research, which typically involves 1,000 or 2,000 people in an attempt to obtain a representative picture of the general population (i.e., the entire electorate). In quantitative studies, relatively large samples are often used to test previously established hypotheses, and the resulting data are summarized.

Quantitative research
This type of research gathers numerical data and analyzes it via mathematical methods.

Qualitative research uses, amongst others, text-based data that must be interpreted, i.e., qualitative research examines the individual case (Saunders et al., 2019, pp. 179–180). In qualitative research, the result is a conclusion obtained from an interpretative process. Because interpretation is by nature subjective, the results of such research are up for debate and their validity can be questioned. Therefore, qualitative research typically does not aim at representativeness. Rather, specific participants are selected, with the expectation that they can provide in-depth insights into the subject being researched. Due to the very complex process of data collection and analysis, it is usually only possible to work with small samples when performing qualitative research. For example, researchers might be interested in the reasons why people do not vote in an election. In order to explore deeper motives here, interviews with a few participants are conducted where they explain in depth why they do not vote in elections. The aim is to obtain useful and rich information. In addition, one might want to ask why job turnover in certain industries is higher than in others. Here, interviews capture motives and underlying causes and are discussed in confidentiality; the information gathered from these interviews might otherwise not have been obtained with a standardized questionnaire.

Qualitative research
This type of research gathers non-numerical data and analyzes it through meaning interpretation.

Quantitative research is sometimes said to mean that mathematical accuracy does not automatically mean "correct" with regard to research questions. This is how, for example, mistakes are made confusing correlation with causality (Reed, 2005 as cited in Saunders et al., 2019, p. 148). A light-hearted example can be that the number of human births per year in Germany and the number of storks (which are said to fly the babies to the expecting parents' homes) living in Germany have both declined considerably over the last 70 years. If the biology department was not paying attention, they might immediately conclude that there are fewer children in Germany today because there are also fewer baby-bringing storks. Here, the numbers actually **correlate**, but the **causality** is not given.

Correlation
If two phenomena are related in some way or have some sort of connection, they are considered correlated.

Causality
If one phenomenon is found to affect or influence another, they are said to have a causal relationship.

Qualitative research is often assumed to be pure storytelling, and, in this respect, one often speaks of anecdotal evidence. This means that something is recognized as a research result because it has been said often enough. This is associated with the accusation that scientists using qualitative research only find out what they assume (Diefenbach, 2009). If, for example, female participants in middle management of a company are asked about the “glass ceiling” (the phenomenon in which women are discriminated against when seeking leadership positions), it is more likely than not that the interviewees actually confirm the existence of the glass ceiling in their statements. Here the result is expected—the knowledge gain remains small because it is obvious that the researchers asked questions for which they already knew the answers. It is therefore particularly important in qualitative research to reflect on one's own research process and to analyze it critically.

Looking at these two very different research strategies—quantitative and qualitative research designs—the ongoing scientific debate about which approach is “better” is easily understood (Bell & Waters, 2018, pp. 23–27). In fact, one can argue that there is no “better approach” per se, but that either research strategy is ideal depending on the particular research project. For example, for large samples, a research object that can be easily quantified, and the requirement for generalizability (i.e., representativeness) of the results would definitively lead to quantitative research. The desire to gain deeper insights into a little-explored field with more text-based data makes qualitative research design seem reasonable. Researching employee satisfaction in a company would probably be better determined quantitatively, whereas conflict and trauma within teams would be investigated more qualitatively. The sciences and humanities understand that these approaches do not contradict each other, but can also be used together in a process called triangulation.

Types of Scientific Reasoning

The second research decision concerns scientific reasoning. Here a distinction between induction and deduction is drawn, both closely linked to underlying research parameters and the decision for a qualitative or quantitative research strategy (Saunders et al., 2019, pp. 152–155).

Induction
This is a process by which individual cases are used to derive a generalization.

Induction first looks at an individual case or a few cases and draws general conclusions (Saunders et al., 2019, pp. 154–155; Veal, 2018, pp. 48–51; Sheppard, 2004, pp. 49–52). If one has no knowledge of car brands and their specific characteristics, but a Ferrari passes you on the road, you might conclude that Ferraris are always very fast cars. Although you may be familiar with this car maker, it could have been pure coincidence and all other Ferraris are totally underpowered, slow vehicles. Induction is therefore often used to develop a theory from a few single cases, which one could test further, with more observations, so as to achieve a more solid conclusion.

Deduction
This is a process by which a general premise is applied to specific, individual cases.

Deduction is the opposite of induction and moves from general conclusions to individual cases. Typically, a theory is used, and this theory is tested on a case-by-case basis (Saunders et al., 2019, pp. 152–154; Veal, 2018, pp. 48–51; Sheppard, 2004, pp. 49–52). If the theory states that all Ferraris are particularly fast cars, then it follows that a Ferrari standing on the street in a residential area must be a fast car. To be on the safe side, the theory (all Ferraris are fast cars) would be investigated by testing the Ferrari. If the car proves to be fast, the theory would be confirmed; if the Ferrari is slow for any reason, the theory would be refuted or falsified.

Quantitative research usually uses deduction. Induction is more common in qualitative research; however, there are times when deductive approaches are taken.

Type of Data

When it comes to the basic decisions of research, it is important to consider the different types of data. A distinction is made between **primary and secondary data** (Veal, 2018, p. 52; Rea & Parker, 2014, pp. 4–5). Primary data are generated for the purpose of the current investigation. Therefore, when interviews are conducted, the interview records are considered primary data. The same applies with regard to participants' individual answers on standardized questionnaires. The primary use of such data is for the researcher who collected the data.

Primary and secondary data
Primary data are generated for the investigation, while secondary data were originally collected for a purpose outside the research project.

Secondary data are data originally collected for a different purpose, which is then used for a new and different investigation (Saunders et al., 2019, pp. 316–318; Tantawi, 2021). For example, if a market research institute uses economic data to draw an absolute comparison between two countries, and a different research group uses the same data to analyze whether there is a correlation between religious affiliations and economic data of various countries, the data are known as secondary data.

Type of Research to Be Carried Out

How research is implemented is also part of the fundamental decisions of research. Here, a distinction is made between **experimental** and non-experimental investigations (Creswell & Creswell, 2018, pp. 11–13). One possibility of non-experimental investigation is the "field-research". This type of research is carried out "in real life" (Arrington, 2021), e.g., interviews with managers at their workplace. It is particularly important for observations to study participants' behavior in real context and then analyze it. Experimental research means creating a controlled environment in which investigations will be carried out. The advantage here is that the environment is controlled and any interference can be ruled out (Stoica, 2021). In marketing, for example, laboratory supermarkets (i.e., those that only simulate the shopping situation) are used to investigate whether and how customers can be encouraged to buy certain products. The fact that a shelf with sweets is typically found at the cash register is probably the result of experimental studies in which it was established that children like to "motivate" their parents to buy another chocolate bar while waiting in the checkout line.

Experimental investigations
These types of investigations take place in a planned environment.

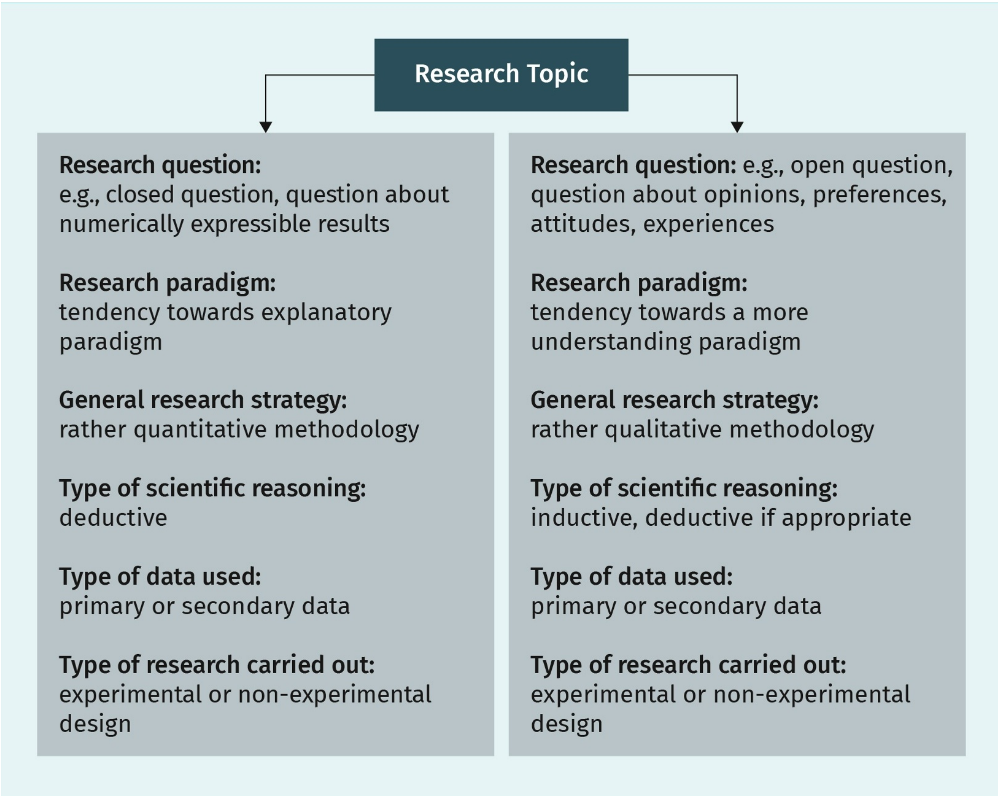
The research decisions mentioned here represent some of the cornerstones of research upon which data collection and analysis can be built and planned.

1.4 Impact of Scientific Paradigms on Research Design

The essential assumptions about research have a direct impact on research design (Saunders et al., 2019, pp. 128–132; 172–175). Since research paradigms allow conclusions to be drawn about the underlying assumptions of research, this results in a logically stringent sequence of research design. Research is a process that one follows, much like a formula. Accordingly, it must be possible to establish, critically reflect upon, and justify the research approach.

The following graphic provides information on the connection between research paradigms and research design.

Figure 2: Connections between Basic Research Decisions



Source: Created on behalf of IU (2020).

The figure illustrates that with the formulation of the research question and the research object, a large part of the research decisions have already been made, or at least a specific research strategy appears probable. The following two examples provide further information.

Example: Introducing rent control

The government plans to introduce new legislation to limit rent increases (rent control) and institutional landlords will be questioned about their attitude to such regulation and the associated challenges. The research question is formulated:

> How do institutional landlords assess the introduction of rent control in relation to their business model?

From this question one would lean more toward an understanding research paradigm, since it is clearly a question of personal assessment. This offers a qualitative research strategy that focuses on the analysis of text-based data. The form of scientific reasoning may tend to be more inductive, as a few cases are considered and theory is generated rather than tested. If experts from rental companies should be interviewed, primary data would be generated. If the survey were to take place at the workplace of the interviewees, it would be a non-experimental research design.

Example: Kids and chocolate

The goal is to discover whether parents who shop with their children buy more of the chocolate bars prominently displayed at supermarket checkouts than adults who shop alone. The following research question is formulated for this purpose:

> How does buying behavior change with regard to chocolate bars offered at supermarket checkouts when adults go shopping with their children?

Here one can assume a rather explanatory paradigm, since number-based data are most likely used and a representative research result is the goal. In this respect, a quantitative research strategy can be assumed and thus also a deductive—in this case theory-tested—form of scientific reasoning. If video recordings of a laboratory supermarket in which a research institute originally observed the buyers in relation to a different issue were used, secondary data would now be employed when using the recording for this research. The setting corresponds to an experimental research design.

Research should be structured in categories so that the readers seeking results can understand and comprehend the process and outcomes. This still leaves room for criticism and discussion, both of which are highly important as this is the base upon which scientific progress is built.

SUMMARY

Science and research differ to a considerable extent from the everyday knowledge and experience of an individual. In this unit, scientific theories have been introduced in order to provide initial insight into available research paradigms. These theories serve as a starting point for scientists and academics to make important research decisions and to choose which research design works best for their empirical and/or

experimental inquiry. With the help of practical examples and scenarios, an application-oriented introduction to this often very abstract and challenging subject is provided.

UNIT 2

PRACTICAL APPLICATION OF GOOD SCIENCE

STUDY GOALS

On completion of this unit, you will have learned …

- about the importance of ethics in research.
- how to verify and evaluate the quality of scientific studies and research.
- when using a non-disclosure agreement and affidavit is required.
- what role spelling and structure have in academic writing.
- how to identify a topic and continuously narrow the research question.
- how to develop a research question and use it to scaffold academic work.

2. PRACTICAL APPLICATION OF GOOD SCIENCE

Case Study

Maike is an ambitious employee in a large company. In order to advance further in the company, she has decided to go back to school part-time to pursue a bachelor degree. While the content of her studies interests her, she quickly realizes that she has difficulties with academic work. The first written assignment is already torturous for Maike, making her uneasy about her bachelor thesis, which requires an empirical study. She will need to conduct an investigation and either develop or test hypotheses, depending on which research paradigm she chooses. Eventually Maike decides to test a hypothesis empirically for her second research essay, which is due soon. She chooses a qualitative approach without knowing whether it is suitable for her research question. Ascertaining the research design and data collection become quite complicated, and, after a third failed interview, she is about to give up altogether. The pressure is mounting and the due date is fast approaching. In her distress, she invents the missing interviews and lets a fellow student—Joana—do most of the analysis. To thank Joana for her help, she invites her to a popular music concert. Maike submits her essay on time and states in the affidavit that she carried out the research independently without the help of others—only making mention of her acquaintance Joana, in the acknowledgments. Despite the weak scientific basis of the work, she is compelled by her professor to publish the results in an IU paper, making them available to the broader scientific community. Maike now has moral dilemma because she manipulated her data. She is afraid that a larger number of critical readers will increase the possibility that someone will discover her deception, especially when she learns that her work will be scanned by plagiarism software.

In the specific case of Maike, many questions can be asked with regard to good scientific practice.

- Why is honesty important, not only morally, but even existentially, in scientific research?
- What drives researchers to manipulate or falsify results?
- How can an outsider judge whether a study is trustworthy and qualitatively reliable?
- How can students avoid such early defeat and demotivation with academic writing?

It is important to show students how to apply and implement good scientific practice from the very beginning. The following content on how to implement good scientific application should be practiced with all IU-specific examinations, and in particular with data protection, affidavits, correct spelling and formatting, and topic identification or differentiation.

2.1 Research Ethics

When it comes to ethics in research, many people likely think of crimes committed against groups of people in the name of scientific progress. Throughout the history of modern science, people who did not have the power or the knowledge to defend themselves were exploited in experiments in the name of scientific or medical progress in a way that today are considered immoral. Especially in medicine, which starts with the ethos of healing people or at least alleviating suffering, such trials are a particular outrage, as with the infamous Tuskegee study in Alabama, USA from 1932 to 1972 (Benedek, 1978). This study investigated the natural course of syphilis; however, the African-American male patients were neither told what they were suffering from nor given access to medication, causing many to die in agony.

Ethical issues naturally arise in other disciplines as well. In the economic and social sciences, for example, protecting study participants' data and personal information is becoming both ever more important and challenging. As gradually the private and the public with increasing social media use is mixed, it is becoming very difficult for researchers to effectively protect study participants from the misuse of data collected (Canadian Institutes of Health Research et al., 2018; Saunders et al., 2019, p. 259).

However, ethical issues need to be taken into account right from the beginning of any research project and need to be present with every step of a work that claims to be scientific (Saunders et al., 2019, p. 252). Why? Postulating scientific results means claiming truth. The concept of truth—also scientific truth—is highly complex and books written on this subject fill the library shelves of all major universities (Ghins, 2017). Put simply, reliability and accessibility of scientific results form a basis for the work of all scientists worldwide—the scientific community. Thus, it is essential that scholars respect academic honesty and can be trusted. It is this honesty that encompasses all the steps necessary for scientific insight and discovery. From topic selection to publication, incentives and pressure can become so high that researchers could find themselves guided by motives other than those grounded in the pursuit of science and truth.

Robert King Merton, an American sociologist and science theorist, formulated ethical norms that are still regarded as the foundation for science (as a system or institution), even as they are constantly adapted and redefined over time (Bucchi, 2015).

According to Merton, these standards are also referred to as **CUDOS principles** (Merton, 1973) and include the following.

CUDOS principles
These principles describe the four basic ethical norms in scientific research.

1. Communism: Scientific knowledge does not belong to the individual researcher, but rather to the scientific community and society as a whole, which carries and facilitates all research. Scientific results can only be recognized when published and made available to the community.
2. Universalism: Scientific statements are valid regardless of the social class, religion, race, or other characteristics of the researcher. Scientists should only be judged on the basis of their scientific performance.

3. Disinterestedness: Scientists and scientific institutions may only act for the sake of scientific progress and not for personal reasons (e.g., career, political opportunity).
4. Organized skepticism: Scientific results should not simply be believed, but must be reviewed by the researcher and by the scientific community; only then can scientific truth be affirmed. This control must be anchored in scientific institutions and in the broader scientific system.

Robert Merton (1973) postulated these norms not so much for the individual researcher's morality—it was clear to him that scientists are people just like everyone else—but rather for the institution or system of science, which, in his opinion, cannot exist without these foundational norms.

It is not difficult to imagine in how many places these principles are challenged today: authorship in the digital age; the increased competitive pressure among researchers to publish quickly and frequently (how, for example, can the scientific community still accurately check the flood of publications?); and influential persons and institutions ("gate keepers"; Bucchi, 2015) who determine who gets what position or which research grants, etc. The list of breaking points for honest research goes on and on. After a few blatant, public cases of scientific misconduct, almost all major national and international research societies, universities, and other scientific institutions have created for themselves a reputable set of standards for scientific work, its review, and implementation (for the U.S. and Canada, as well as some international guidelines, see the compilation of The University of British Columbia, 2019 or Saunders et al., 2019, pp. 254–255).

Saunders et al. (2019, pp. 257–259) considered ten ethicals principles, that occur across many different approaches for research:

- integrity, fairness and open-mindedness of the researcher
- respect for others
- avoidance of harm (non-maleficence)
- privacy of those taking part
- voluntary nature of participation and right to withdraw
- informed consent of those taking part
- ensuring confidentiality of data and maintenance of anonymity of those taking part
- responsibility in the analysis of data and reporting of findings
- compliance in the management of data
- ensuring the safety of the researcher

One could perhaps formulate a brief summary of the international efforts toward scientific honesty in this way: anyone who produces sloppy work, deliberately conceals, falsifies, or invents results, or does not appreciate employees, destroys the possibility of science. If the scientific community and its institutions cannot establish rules, verify compliance, and enforce them, not only will the reputation of science be damaged, but science itself will no longer have a chance to pursue the discovery of universal truths.

2.2 Evidence

After exploring ethics in research, it is necessary to examine the role of evidence in science (Elsevier Author Services, n.d.). However, first, a few words about the term "evidence". Evidence is a much-discussed term in philosophy and science (philosophers such as Ludwig Wittgenstein and Edmund Husserl, for example, could be mentioned here). Generally speaking, evidence is "the available body of facts or information indicating whether a belief or proposition is true or valid" (Lexico.com, 2019b).

It is important to mention that not every study meets the standards of good science (Elsevier Author Service, n.d.). Studies show differences regarding the quality of research design, implementation, and review. So, how can one determine whether or not an experiment is reliable and qualitatively acceptable? This question may surprise some, as it is generally assumed that scientific studies always provide new and reliable results. Unfortunately, this is not always the case. This makes it all the more important to subject studies to a review. In order to assess whether a study provides reliable data, it is important to determine the reason why the experiment was conducted and what question it investigated. This may sound trivial, but it is crucial in order to determine whether the study is able to thoroughly answer the research question. Subsequently, the methodology of the study should be considered, in particular whether the methodology used was appropriate to answer the research question, whether it was carried out properly, and whether there were systemic errors (bias) that could distort the results.

The following questions can help to evaluate studies:

- Is the study design suitable to answering the research question?
- How were the participants approached and selected?
- Who or what was included/excluded and why?
- Did the researchers describe the procedure and the results completely and comprehensibly so that the study could be repeated and reviewed?
- Were the samples large enough and the number of experiments frequent enough to answer the research question?
- Was the study conducted long enough?
- How many participants or testing arrangements/experiments were eliminated or unsuccessful during the study and why?
- How many participants or experiments could no longer be investigated during the period post-study (for follow up) and why?
- Was the study conducted by an independent institution or by the private sector?
- Are the samples and number of tests performed representative and sufficient to establish conclusive evidence?

In addition to this fundamental assessment and evaluation of a research study, there are levels of evidence, or classifications of evidence, that provide information on whether research is of high quality and whether its results can be scientifically communicated.

"Evidence class" is a term mainly used in evidence-based medicine to describe and categorize the formal and substantive quality of a study. It describes a hierarchy of evidence. The scientific significance is thus evaluated with the help of evidence classes, or levels of evidence (Elsevier Author Services, n.d.). The five levels of evidence hierarchy are (Mantzoukas, 2007, p. 217):

Meta-analysis
This method tries to statistically summarize early research projects on a certain topic. It is a summary of many sources of primary data.

Randomization
In this procedure, participants are assigned to different groups in an unplanned manner.

- Level 1: Evidence from at least one **meta-analysis** of RCTs (Randomized Controlled Trails)
- Level 2: Evidence from at least one well-Conducted RCT
- Level 3: Evidence from controlled research without **randomization**
- Level 4: Evidence based on research without experimental study
- Level 5: Evidence based on opinions/reports of recognized authorities

Definition of evidence levels or classes may vary between organizations and differ between specialties, according to the clinical question being asked (Murad et al., 2016; Burns, 2011).

2.3 Data Protection, Affidavit, and General Legal Information

In many research areas, working with personal data is inevitable, whether it is a written data record, video recordings, or images. All of these possible empirical sources and information are subject to data protection regulations. Those working in scientific research must therefore know and comply with the applicable regulations, consistently implementing them in their scientific practice (Saunders et al., 2019, p. 276). In the following, aspects of data protection that must be taken into account in academic writing will be explored.

Collection of Personal Data within the Scope of Surveys

If the survey requires personal information, i.e., if the respondents can be identified, various data protection measures must be put in place (Saunders et al., 2019, pp. 276–278). This is also the case even if a name or other personal data are not identifiable and are replaced or redacted, making it very difficult to identify the person in question; in such a case, one speaks of pseudonymity. Regardless, pseudonymous surveys must be treated just like surveys containing personal data.

Examples of such guidelines include

- data minimization. Collecting too much personal data should be avoided. (What data are absolutely required for the purpose of the survey and how do they have to be for the research purpose?).
- legality. The collection and processing of personal data may have a legal basis. If there is no legal permission for the processing of data, the consent of the interviewee must be obtained.

- deletion of data. Personal data must be deleted as soon as they are no longer needed. For scientific surveys, the data must be kept for ten years, even if the survey is part of an examination. At the end of the ten-year period, the data must be deleted.

When disseminating or publishing the results of scientific surveys, all personal information should be removed beforehand. For example, if a student conducts a survey that is part of a research essay, the survey participant data should be pseudonymized before the essay is submitted. This means that names must be replaced by pseudonyms or completely redacted. No names should be visible so as to ensure the anonymity of the interviewee.

If no personal data is collected and/or used in a survey, no data protection consent has to be obtained. Personal data is not collected or used if the data is completely anonymous, i.e., if it is not possible to draw conclusions about the person or an attribution to a person.

Copyright Laws

The use of graphics and photos in academic work can be tricky, especially when figuring out who owns the graphic or photo in question and if consent is given for its use in text and on other platforms, such as on social media. In order to use such images, one must follow the law of the country. The regulations for using photographs in Germany, for example, are clear: all photographs are protected by copyright (§ 72 UrhG).

The following provides clarification on the usage of graphics and images when submitting an academic paper in Germany, respectively on a university located in Germany.

Use of images (without personal reference)

Images, graphics, and maps that have been previously published (e.g., available online) may only be used if the work references them. This means that there must be a a relationship between the image and the text (Section 51 No. 1 UrhG). In addition, the creator of the image needs to be cited. The Citation Guidelines show how to correctly cite images.

Questions:

1. Is this image necessary?
2. Does the idea remain valid without the image?
3. If a trademark (image, text, water mark, etc.) is depicted in the image in question, the following applies: In principle, trademarks (logos) are also protected by copyright. It must not appear that there is a relationship between the trademark proprietor and the user if there is actually none. A copyright infringement of a trademark is excluded if it is only an “insignificant accessory” in the overall image. The decisive factor for this is that the object protected by copyright and trademark law cannot be the main design of the image (e.g., a company sign in the image of a large shopping street). Icons are also protected by copyright. If icons are used, the creator of the icon should also be named, unless the icon is in the public domain.

Concerning media, Microsoft grants a license for its products such as Word and PowerPoint in projects and documents to copy, distribute, and display media elements (images, clip art, animations, sounds, music, video clips, templates, and other types of content) that are part of the software. The content contained in those programs may therefore be used for handouts and presentations provided that they are reproduced for a limited number of participants and not used commercially. This also applies to research essays. However, the use solely for illustration purposes is not permitted and always requires the consent of the copyright holder.

Images from social media are also protected by copyright. Pictures of people as well as of landscapes from social media platforms are copyrighted and may only be used with permission. In the case of pictures in which people are depicted, in addition to the consent of the author/photographer, the people depicted in the image themselves must also consent to their image being used. Such consent shall not be subject to any condition and must contain detailed information on the data used, the purpose of the data used, the storage period of the data, and whether disclosure to third parties is planned. Due to the large amount of information contained in such consent, it is imperative that it be obtained in writing.

Screenshots are also treated mostly like images where many of the same considerations apply. It must serve to clarify the text and cannot be used simply as an “insignificant accessory”. Screenshots of websites and videos are to be treated like image citations. In addition to the image citation, one should obtain the consent of the author or person depicted.

Non-Disclosure Agreements

In order to protect data and information when pursuing scientific work, there is the option to use a non-disclosure agreement. This is particularly helpful if a scientific paper is written in cooperation with an external cooperation partner or with a company. Each study can be provided with a non-disclosure or confidentiality agreement. The latter stipulates that the work—without the express consent of the company and the author—will not be made available to third parties, with the exception of the supervising lecturers and authorized members of the audit committee. The non-disclosure agreement prohibits publication for a specific period of time in an effort to protect sensitive data and research results. At the same time, the scientifically necessary publication of research results can be partly delayed or completely prevented (if the results do not appear optimal), which repeatedly fuels criticism of privately financed research. Therefore, a compromise must always be found between the interests of the company and the interests of the university or the scientific institution and community. Violation of a non-disclosure agreement may even result in criminal prosecution.

Affidavit

An affidavit must be included in every academic work. With this declaration, the author confirms that the work is independent and was not produced with outside help or through the use of outside intellectual property. It is also confirmed that no sources other than those stated were used. The affidavit must be included in the work. It is not submitted as a separate document but forms the last page of the written elaboration (after the annexes). A missing or unsigned affidavit leads to failure of the examination. Unlike many other universities, the declaration at IU is filed electronically before submission. The declaration does not have to be included in the work itself (except for the thesis—further information under Information Thesis in General in myCampus).

EXAMPLE AFFIDAVIT

I hereby swear that I have done this work independently and without the use of outside help. The sources and tools I used are clearly marked as such and referenced at each instance in the text. This work has not yet been submitted in the same or a similar form to any other examining authority. I agree that the work will be checked for plagiarism with the help of a plagiarism detection service.

____

Signature

2.4 Spelling and Format

A research paper must meet certain formal criteria which serve to ensure legibility, transparency, understanding, and clarity (Saunders et al., 2019, pp. 731–739). The excessive use of personal pronouns as “I” and “we” should be avoided in academic writing (Saunders et al., 2019, p. 737). If the researcher takes an active part in the research process, moderate use of personal pronouns could be reasonable. Language should be gender-neutral, non-binary and not discriminating. More information is found in the IU guidelines for gender-inclusive/sensitive language available in myCampus.

The spelling (orthography) of each work should be adapted to meet the current standard and must be applied consistently. Spelling and typing errors should be avoided by reading the work several times, and one can even use third-party reviewers, before submission. Students whose native language is different from the one used in the examination should have their work proofread by a trained native user for the review of language, style, and understanding.

The paper is preceded by a title page on which the author (complete with name, postal address, email address, and student number), corresponding event (course, lecturer, semester, and title), and topic and type of work (written assignment, research essay, bachelor thesis, etc.) are noted.

Apart from the title page, all pages must be numbered. The pages before the body of the text (e.g., title page, table of contents, list of tables and abbreviations) should be numbered in Roman capital letters (I, II, III, IV, etc.), with the page number not appearing on page I (title page). The pages of the text part are numbered with Arabic numbers (1, 2, 3, etc.), beginning with page 1. This is usually the bullet point “1 Introduction”. These page numbers are continued to the end, i.e., also through the appendix. The desired position of the page numbers is centered at the end of the page. Further form specifications can be found in the Exam Guide in myCampus for every type of examination.

Ideally, illustrations and tables should be created by the author; previously created graphics and images should only be used when absolutely necessary.

Images, sound, and video material are subject to copyright and must be identified by citations. Rules for correct citations are found in the general citation guidelines.

2.5 Identification and Focus of Research Topics

Before a topic for a scientific paper is decided on and research is started, one should check whether there are certain requirements for that topic area. This applies to all types of academic work.

The following questions should be answered “yes” if the topic is chosen freely.

- Is the subject interesting to me?
- Can I imagine dealing with the topic intensively for a while?
- Do I have previous knowledge of the chosen topic?
- Is the topic of interest important for my further studies and career goals?
- Does the topic meet the requirements in terms of scope?

Topics which may prove unsuitable include those that

- have been extensively covered (“trending”) and for which there is a wealth of existing literature.
- represent a strong personal connection, such that scientific discussion and objectivity might become difficult.
- are still explorative, i.e., for which there is little or no literature available.
- require the use of sources that are difficult to find or completely unavailable.
- require the use of sources that are too demanding technically or linguistically (foreign languages).
- require the use of methods that are not mastered or are unavailable within the given framework.
- are too abstract, making a practical scientific approach difficult.
- require elaborate empirical approaches.

It is not always easy to answer these questions. For example, whether a topic is extensively researched (or not sufficiently researched) and unsuitable for one's own research work, does not necessarily become apparent after the initial literature search. It is also not always easy to recognize which methods are most suitable for dealing with a topic (Saunders et al., 2019, pp. 26–29). Ideally, interesting topics and questions arise continuously throughout the studies. It would makes sense to solidify topic identification and avoid abstractions.

Once the working title of the topic is named, the research question to be answered is derived from it (Saunders et al., 2019, p. 42). Depending on the scope of the scientific work, it is also common to formulate two or three research questions related to a topic. These questions help narrow the topic and serve as a common thread throughout the work, and also help with the preparation of the outline.

These are examples of poor titles:

- Leadership in the 21st Century
- Work 4.0
- Digitization in the World of Work

These are examples of better titles:

- The Model of Transformational Leadership in the Energy Industry: Innovation or Trend?
- Influence of Digitalization in the Service Industry Using the Example of Call Center Software in the Insurance Industry
- The Effect of Digitalization on the Travel Behavior of Generation Z—An Overview of Current Applications in the Tourism Industry

Using thematic boundaries and constraints can help to determine which aspects should be addressed, which should not, and why. The following example considerations can be used as criteria to narrow the topic:

- time
- geography
- institutions
- groups of people
- sources
- people or authors
- aspects of a particular discipline
- theoretical approaches or explanatory concepts
- theoretical or explanatory approaches according to experts
- specific, selected aspects

2.6 Research Question and Outline

The research question is an integral part of the outline of a scientific work and is embedded in the introduction (Saunders et al., 2019, pp. 714–715). The entire outline must follow a common thread, which develops according to the research question being asked. The following image is that of a funnel. The research question in this case is the funnel (with built-in filter), which decides the classification points and facts to be included. Everything that can be processed within the outline of the work and contributes to the assessment of the research question must be included. The outline should demonstrate how the research questions is executed. It must reflect a logical development of the work as well as provide an overview of it.

Figure 3: Funnel Function of a Research Question

Source: Created on behalf of IU (2020).

Examples of research questions include the following:

- What influence does a manager's leadership style have on employee health?
- Which evidence-based models are available to application-oriented research on healthy leadership in companies?
- What research needs arise from the current state of research on healthy leadership?

These examples of research questions could all be chosen and developed under the title "Leadership and Health—A Challenge for the Mechanical Engineering Sector".

Outline

In the outline, all outline points are listed, if necessary with sub-points. If a subdivision of an outline point is necessary, it must contain at least two sub-points.

Table 2: Structure of a Scientific Outline

Poor	Better
1. Introduction	1. Introduction
1.1 Problem Statement	1.1 Problem Statement
2. Theoretical Foundation	1.2 Objectives and Research Questions
	1.3 Structure of the Work

Poor	Better
	2. Theoretical Foundation

Source: Created on behalf of IU (2020).

As a rule, the work should be divided into a maximum of three section levels (1. Main Heading, 1.1 Section, 1.1.1 Subsection). The sections and subsections must be identical in title and numbering as those mentioned in the text. Because section headings, and the corresponding page numbers, often change during the course of writing, it is advisable to work with corresponding font formats (Heading 1, 2, etc.) from the beginning and create an automated table of contents.

The following parts occur in every scientific paper:

- table of contents (outline with page numbers),
- text part (consisting of introduction, body and conclusion),
- bibliography.

Other indexes, such as list of figures, list of tables, and list of abbreviations, help the readership to find pertinent additional information. The lists of tables and figures as well as the list of abbreviations are listed in the front part of the paper, i.e., between the table of contents and the text part. A list of tables must be listed for three or more tables and a list of figures for three or more figures. Another optional part is the appendix. It serves to present information that is too detailed for the text part, but important for its understanding. This can be the original copy of a survey, large tables, or scanned materials and transcripts of in-depth interviews. Each appendix must be labeled as an “Appendix” with an appropriate label. For example: Appendix A, Appendix B, etc. Appendix pages are numbered but are not included in the specified page count of an examination paper. Each appendix should be referenced in the paper.

Only the individual sections in the text of the paper are numbered in the outline; the other outline components of the paper, such as the table of figures or bibliography, are indicated without numbering. The following basic outline must be taken into account for the text part of a paper (Saunders et al., 2019, p. 714):

- introduction
- body (with theoretical background/literature review, methods, findings/results and discussion)
- conclusion

While it is common to include “introduction” and “conclusion” as chapter titles, for the body of text you need to choose chapter titles that are suited for your specific research topic. However, it is important to internalize the basic structure of a scientific paper. Especially the chapters, possibly with their subchapters, in the body must usually be adapted to the specific content. However, it is important to internalize the basic structure of a scientific paper.

In the introduction, the research question, objective, problem statement, and structure of the work are described, and relevant and interesting facts are provided. It is here that you want to take a look at what might motivate the reader to continue reading (Saunders et al., 2019, pp. 714–715).

The theoretical background serves to classify the entire topic and research question into a scientific, evidence-based research context. It is important to present existing knowledge and its relevance to the research question (state of current research) (Saunders et al., 2019, p. 715). In this context, the stringing together of term definitions should be avoided. Therefore, instead of devoting an entire subsection to the definition of terms, these are explained the first time they are mentioned in the text.

Table 3: Structure of a Scientific Outline

Poor	Better
2. Theoretical Foundation	2. Theoretical Foundation
2.1 Definition of Terms	2.1 Health in the Context of Work
2.2 Health in the Context of Work	2.2 Leadership as an Influencing Factor on Health in the Workplace
2.3 Healthy Leadership	2.3 Transformational Leadership as a Model

Source: Created on behalf of IU (2020).

The method, or research design, describes the how and why in an academic work. In the methods section, the decision to use a certain methodology is explained in more detail. The advantages and disadvantages of the chosen method must be critically considered and discussed (method criticism) (Saunders et al., 2019, p. 717). In this part of the paper, the choosen method and its specific steps are described; e.g., if a systematic literature search was conducted or which empirical process was chosen (qualitative or quantitative).

In the findings/results section, the own results, which were generated with the applied method, are presented neutrally, i.e., without evaluation and interpretation. In literature-based work, the results of the literature review are presented. If empirical methods are used, the results of the empirical investigation are presented in this part (Saunders et al., 2019, p. 718).

The discussion serves as the final reflection and comparison of the findings against the background of the current state of research, and, where appropriate, political or social developments. Once again, one must reiterate the conclusions to the research question in a relevant and lively way. It is also possible here to describe a model for best practice or to provide recommendations for further action. The identification of further research can also be part of this discussion (Saunders et al., 2019, pp. 718–719).

Finally, the conclusion summarizes the entire paper and may also include the subjective opinion of the author (Saunders et al., 2019, pp. 719–720).

SUMMARY

Good scientific practice places many demands on the author of a scientific work. This unit provides a complex profile of the requirements that students are advised to start dealing with at the beginning of their studies to achieve systematic success. Thus, questions of scientific integrity are not only important with regard to the recognition of one's work, but are also prerequisites for science. This is one of the reasons why affidavits are required for scientific papers.

Furthermore, when doing research and writing scientific papers, the examination of data protection and copyright laws is relevant. However, pragmatic guidelines on the structure, form, and outline of a scientific work must also be understood in terms of their form and scope.

UNIT 3

RESEARCH METHODS

STUDY GOALS

On completion of this unit, you will have learned ...

- the difference between data collection and data analysis.
- the essential characteristics of quantitative methods.
- the essential characteristics of qualitative methods.
- how to explain the quality criteria of research.

3. RESEARCH METHODS

Case Study

It's a week before the parliamentary elections. Simon and Maike are very interested in politics and world affairs, and both have been politically engaged for years, albeit with different convictions and political orientations. This makes such political and historical events exciting for them both—not to mention the rich discussions. One week before the big election, they watched a political debate on television. Directly after the end of the broadcast, a large opinion research institute conducted a survey on party preference ("If elections happened today, which party would you vote for?"). The moderator explains that 1,000 representative participants were surveyed by phone, and the result of the survey will be announced on the next show.

According to Simon, "facts", as presented by the opinion research institute, are unclear and imprecise in many respects and leave much room for interpretation. He explains to Maike that he can hardly believe that the representative sample of 1,000 people accurately reflects the electorate. He is curious on how the sample was selected and the ratios for gender, age ranges, and educational backgrounds. From his own professional experience, he knows how difficult it can be to reach target groups for opinion research. In addition, Simon raises other questions: Were both voters and non-voters really questioned here? How meaningful is such a hypothetical question in the context of a quantitative survey? Would it have made more sense to supplement the survey with an open question on the reasons, i.e., to combine quantitative with qualitative methods? These many questions result in Simon and Maike having an in-depth discussion on empirical research and the associated instruments that lasts long into the night. Seeing how close one's own life is to application-oriented research is quite exciting.

3.1 Empirical Research

When doing research, a wide variety of methods are used (Veal, 2018, p. 132). Depending on the research objective and the underlying research paradigm, qualitative methods or quantitative methods can be used (Saunders et al., 2019, pp. 175–178). It is also possible to choose a mixture of qualitative and quantitative approaches, often referred to as methodological triangulation (Saunders et al., 2019, pp. 181–184). Since qualitative and quantitative research methods have specific advantages and disadvantages, triangulation often serves to compensate for the disadvantages of the respective methodology and should thus lead to a more meaningful research result.

When considering the different methodological approaches, it is important to distinguish between data collection and data analysis (Saunders et al., 2019, pp. 176–180, Veal, 2018, p. 47). Data collection has the purpose of generating the necessary data. Data analysis aims to evaluate the available data, which can then be interpreted accordingly. Since the various research methods have both advantages and disadvantages, it is important to think about the quality of an investigation. For this purpose, scientific quality criteria are used, which differ according to the type of research (Saunders et al., 2019, pp. 213–219). These quality criteria not only point to the research quality achieved, but also show the extent to which researchers are able to critically reflect on their own approach.

When thinking about the opinion poll discussed above, it is obvious that quantitative research was carried out because the result is based on figures and shows percentages. Moreover, they used a representative sample, although no details are presented that allow to assess the degree of representativeness. The survey was conducted by phone, although it is not known whether landlines, mobile numbers, or both, were included. Furthermore, data analysis has been carried out in a simple statistical way—also referred to as descriptive statistics. With regard to research quality, it could be noted that in the phone surveys the interviewer could have an influence on the results through his or her way of speaking or questioning. For some participants it might be embarrassing to reveal their political preference in a phone survey. Finally, the question could be raised as to whether calling late on a Sunday evening might falsify the results, especially since it remains unclear whether or not only landlines and/or mobile numbers were called. With landlines, for example, the problem is that only a few young people still have them, and that it is more likely that they are out on a Sunday evening rather than at home in front of the TV. A critical attitude toward research questions and their implementation is an essential prerequisite for good empirical research.

3.2 Literature Reviews

Not all academic papers present findings from empirical findings; it is also possible to do a literature review (Gülpinar & Güclü, 2013, p. 44), i.e., summarizing relevant literature and not conducting a survey, experiment, etc. This means that neither quantitative nor qualitative research methods, nor methodological triangulation are used. Literature reviews comprise a theoretical overview that compiles and evaluates the current state of research into specific research questions, i.e., summarizing research results. These papers, written in the style of a review, are known as quality standards in evidence-based research.

The purpose of a literature review is to present and assess the existing knowledge on a specific topic in a concise way (Gülpinar & Güclü, 2013, p. 44). To this end, relevant literature must first be selected, grouped, systematically processed, analyzed, and critically evaluated. A review is therefore more than a series of summaries of different sources. A literature review is also based on a research question. How many studies and literature sources are needed to write a good literature review cannot be answered in general terms.

As a rule, a literature review, similar to an empirical paper, is divided into an introduction, a body (with theoretical foundation, method part, results of the literature search, discussion) and a final part, the conclusion. Due to the abundance of articles, it is particularly important to clearly document the selection process of studies and literature used. Selecting relevant literature is essential for writing a good literature review; therefore, one needs to specify exactly how the search was carried out (Gülpinar & Güclu, 2013, p. 47; Hiebl, 2021, p. 6).

- Which databases were searched?
- Which keywords were used for the search, and in what form?
- According to what criteria were the sources included or excluded?
- How many sources and literature have been included in the literature survey?

These questions are answered in the methods part of the literature review. For a better overview of the selected literature, a table can be helpful which can also be helpful when describing the results.

Table 4: Example of a Table Providing an Overview of Sources

	Source 1	Source 2	Source 3
Question			
Research Subject / Foundational Data			
Theoretical Foundation, Models, Concepts			
Methodical Approach			
Important Results / Response to the Research Question			
Other			

Source: Created on behalf of IU (2020).

3.3 Quantitative Data Collection

When looking at data collection methods in quantitative research, the following methods come into particular consideration (Saunders et al., 2019, p. 178):

- experiments
- surveys
- observations
- content analyses

All procedures must lead to number-based data (Saunders et al., 2019, p. 178). In addition to scientific experiments, surveys are one of the most frequently used methods of quantitative data collection (Saunders et al., 2019, p. 178). Surveys can be carried out both orally and in writing, whereby the written survey has the advantage that researchers have no individual influence on the results, since they do not come into personal contact with the interviewees (Saunders et al., 2019, pp. 193–196). In addition, written surveys offer the great advantage that very large amounts of data can be processed. In quantitative studies, standardized questionnaires are most frequently used, allowing the participant to choose from multiple choice answers. Oral surveys are also used in quantitative data collection with one example being election surveys by phone, which aim to collect data as quickly as possible.

Observations are often used to count certain behavior patterns (Veal, 2018, p. 253). One example would be to observe and count how often consumers with small children buy a chocolate bar at the supermarket checkout. Thus, conclusions can be drawn from the observed frequency of certain behavior patterns, for example on the best possible presentation of merchandise in the supermarket. Text analyses are used in quantitative data collection, for example, to count certain words or word groups from a text. One could, for example, find out how often tabloid newspapers use certain dramatic adjectives in their newspaper articles and whether, and to what extent, this number differs from the one in traditional print media. Content analyses also offer the great advantage that very large amounts of data can be collected and processed later on.

Data analysis then focuses on processing the collected, number-based data (Saunders et al., 2019, p. 564). The analytical methods of quantitative research are often differentiated according to the number of variables used (Cleff, 2019, p. 5). If one variable is taken into account, then it is called univariate analysis method. If, for example, you want to know how salary levels behave in a company, you could determine the average, median, and certain distributions.

If the relationship between two variables is of interest, then the bivariate analysis procedure applies, and if more than two variables are considered, the quantitative data analysis procedures are called multivariate (Cleff, 2019, p. 5). A bivariate data analysis is used to test if, for example, the number of overtime hours worked and the number of sick days in a company are related. A multivariat analysis can, for example, determine which variables affect the choice of equipment package when buying a car; here variables such as age, income, or gender could be taken into account.

3.4 Qualitative Data Collection

Qualitative research methodology requires a close look at data collection. In general, qualitative research is characterized by the collection and analysis of text-based data (Saunders et al., 2019, p. 179). In this respect, the collection of data is mainly concerned with spoken or written texts. Likewise, it can deal with symbols, observations, or artifacts, as long as these are somehow expressed textually. Qualitative data can, e.g., be collected with the following techniques (Veal, 2018, pp. 136, 286; Saunders, et al., 2019, p. 180):

- interviews
- observations
- text analyses

Data collection usually takes place in conversations. These can be individual interviews, expert discussions, or group interviews in which the conversations are either open or semi-structured, meaning that the researcher orients his or her questions to a questionnaire (guideline) (Veal, 2018, pp. 136, 286). In open conversations, the researcher does not use an interview guide, but conducts a conversation on a topic in which both the interviewer and the interviewee express themselves freely. Observations are often made in relation to behavioral patterns, which in turn include conversational situations. Text analysis refers to the interpretation of documents. The subjectivity of the researcher is an integral part of the research process in qualitative research methodology and it is obvious that data collection can be much deeper than is typical in quantitative research. It should also be emphasized that the researcher may ask if an interviewee does not seem to have fully understood a question or if the interview results reveal interesting aspects that need to be explored further. Qualitative data collection often takes place "in the field", for example where the interviewees work. Qualitative data collection is more likely to be applied to relatively small samples because otherwise it would result in large amount of text-based data, making the effort of analysis enormously time-consuming.

Data analysis in qualitative research regularly consists of the interpretational processing of the obtained, text-based data. Different methods are used for this purpose, which will not be presented in detail here, however, the qualitative data analysis methods of Mayring (2000, 2022) and Kuckartz (2014) are resources worth exploring.

It is important for qualitative data analysis to be carried out in a rule-based and comprehensible way. Software is now also available for this purpose, which considerably simplifies the analysis of large text files (Veal, 2018, p. 465). The general aim of data analysis is to classify interview statements (or fragments) into either a given category system or a category system developed from the data itself (Mayring, 2000). For example, employees from different sectors could be asked about their attitudes to certain motivational stimuli. Based on interpretations of interview statements it can be concluded, for example, that it is essentially non-material incentives (e.g., praise, recognition, opportunities for further development) that particularly motivate employees. After these categories have been developed from the analysis of the first interviews, one can check throughout the study whether these categories continue to be confirmed by other interviewees or whether there are more categories to add or a need to modify the existing categories resulting from the interviews. In contrast to quantitative analysis where the response categories are defined in advance, the development of categories in qualitative research is a continuous activity carried out through the end of the interpretation process and is continuously adapted, if necessary.

3.5 Mix of Methods

There are ongoing discussions about which of the methods—qualitative or quantitative—is better (Veal, 2018, pp. 43–44). Quantitative research typically offers a mathematically correct, and therefore understandable, results (Saunders et al., 2019, p. 175). In addition, the aspect of representativeness plays an important role because results in quantitative studies can often be generalized, which proves advantageous (Veal, 2018, p. 53). Recalling the example of representative election research, it not only provides a fairly accurate picture of party preference in the population, but it is also used to forecast election results.

Qualitative research, on the other hand, can be used to more deeply research subject matter and to identify preferences, motives, attitudes, and perspectives (Saunders et al., 2019, pp. 175, 179). In this respect, quantitative investigations are often said to be only mathematically correct and therefore remain somewhat superficial (Guba & Lincoln, 1994, pp. 105–106). Here the question is often asked whether, and to what extent, it is possible to represent complex human actions in a mathematical formula. On the other hand, qualitative research is often said to produce only anecdotal evidence, i.e., to present little more than "telling stories". Often, qualitative research is even denied scientific merit.

An important difference between the two basic approaches is that quantitative research is used to test and either confirm (verify) or disprove (falsify) existing theories. Qualitative methods serve more to develop new theories and are thus typically used in fields in which little research has been done (explorative settings) and in which there are few existing theoretical approaches (Veal, 2018, pp. 135–136).

It becomes clear that there is no one correct method, but that both methodological approaches have their advantages in certain thematic areas and research questions (Bell, 2018, pp. 25–26). Therefore, a widely shared opinion has been established that the two methodologies do not, in fact, contradict each other, but are ideally even used in combination, which happens, for example, in student evaluations or in service satisfaction studies. Very often the participants are first guided through a standardized questionnaire with certain checkbox options. This is the quantitative part of the study. In addition, the participants often get the opportunity to express in writing their personal opinion in an open text box, which is the qualitative part of the investigation. This mix of methods is also called triangulation and offers enormous research advantages (Veal, 2018, pp. 157–158). First, the advantages of both methods are combined and at the same time their disadvantages are balanced. Second, the research result is particularly valuable. While the quantitative part offers generalizable preferences of the study participants, the qualitative part provides further, usually more in-depth, information, which likely was not addressed in the quantitative survey.

3.6 Critique of Methods and Self-Reflection

Research and the results it produces must be justified in terms of quality (Saunders et al., 2019, p. 213). Different quality criteria are used for qualitative and quantitative research methods in order to critically reflect on the quality of the research carried out. The fact that different standards are applied here is due to the various objectives pursued in the two research approaches. Therefore, a comprehensive critique of the methodology is necessary for many academic papers. It shows the researchers' degree of self-reflection and also provides information on what the results can be used for and where their limits lie.

In quantitative research one of the most important questions is whether results generated in a sample are generalizable or representative of the population as a whole (Veal, 2018, p. 53). So if 1,000 eligible voters are questioned about their party preference, is the result

Representativeness
If the result of research completed from a sample provides a viable picture of the characteristics of the general population, it is considered representative.

then generalizable, i.e., valid for the entire electorate? **Representativeness** means that the result of research completed from a sample provides a viable picture of the characteristics of the population. Representativeness is ensured by the type of sample selection and can, depending on the type of population, be statistically represented. In this context, it is important to note that not every quantitative study needs to be representative in order to be valuable. As is the case in many final university exams, a representative sample is not always possible due to time constraints. This does not mean that the investigation is worthless; it only means that there are uncertain conclusions about the population.

Validity
When an investigation measures what it is supposed to measure, the investigation results are valid.

Reliability
When the results of an investigation are replicable, they are said to be reliable.

Objectivity
For the results of an investigation to be objective, they should be independent of the respective researcher.

Furthermore, the quality of a quantitative study is expressed by the aspects of validity, reliability, and objectivity (Saunders et al., 2019, pp. 213–214, 268). **Validity** means that the results obtained reflect exactly what one wanted to investigate. One could, for example, ask whether the measurement of voting intentions actually reflects election behavior. It could be that the respondents, out of general rage, indicate protest parties as election preferences in the survey, but then choose the established parties in the election. **Reliability** refers to the question of whether one would come to the same results if the investigation was repeated under the same conditions. **Objectivity** requires the researcher to have a negligible influence on the processes of data collection and analysis. Electoral research, for example, often uses phone interviews. This procedure reduces the quality criterion of objectivity because the nature of the question could influence the interviewee. An exception of this quality criterion is accepted, however, in order to have results available quickly—an aspect that is naturally important before elections.

Saturation
In data collection, saturation is said to have been achieved if no new findings are generated despite the inclusion of new data.

In qualitative research, other quality criteria are applied (Saunders et al., 2019, p. 216). With relatively small samples, representativeness would certainly not be possible and does not represent the goal of qualitative research. Rather, **saturation** is the aim (Saunders et al., 2018). Saturation is the point at which no new findings arise despite the inclusion of further data records (e.g., further interviews). Even in qualitative research, not every investigation can guarantee saturation. Twenty-five long interviews can quickly lead to the creation of several hundred pages of transcripts. It would not be possible to evaluate this completely in a bachelor or master thesis due to time constraints. Nevertheless, even a non-saturated qualitative study can reveal valuable tendencies.

Further quality criteria for qualitative research are dependability, credibility, and transferability (Lincoln & Guba, 1985 as cited in Saunders et al., 2019, p. 217). Dependability refers to the question of whether data have been evaluated consistently, accurately, and without contradiction. This is ensured by correspondingly clear interpretation rules. In qualitative research, credibility is established by an interpretation process guided by rules in which the results can be checked at any time using the interpretation rules. The transferability can be achieved, for example, by having several researchers evaluate the data and thereby determine the intersections and discrepancies. Where interpretations are congruent, a greater degree of transferability can be assumed.

It is important to describe which quality criteria can be achieved or ensured to what degree for both method criticism and critical reflection in academic work. Any limitations —which are often understandable and acceptable—should also be pointed out. Limitations often result in further research paths that can be pursued in the future. This is a normal and desirable process, since research truly never ends.

SUMMARY

Researchers should always apply and consider the critical examination of research findings in their practice because it is a core competence. Questioning and reflecting on study designs, surveys, presentations of results, and individual research instruments or methods represent existential components in a holistic research process. In order to be able to make these assessments and reflections, it is indispensable to have sound basic knowledge of quantitative, qualitative, and triangulated research methodologies in order to be able to classify and evaluate the quality of research processes.

UNIT 4

ACADEMIC ADMINISTRATION: STRUCTURE, APPLICATION, AND LITERATURE MANAGEMENT

STUDY GOALS

On completion of this unit, you will have learned …

- what plagiarism is and how to avoid it in academic writing.
- why and how you should use research databases.
- which search techniques and keyword searches are optimal for your research question.
- how to reference and cite sources and how to apply academic guidelines in authoring your work.
- what a literature management program can do.
- how to create a bibliography with the help of a literature management program.
- which software and application tools IU offers to support the writing of academic papers.

4. ACADEMIC ADMINISTRATION: STRUCTURE, APPLICATION, AND LITERATURE MANAGEMENT

Case Study

To work on his bachelor thesis, Simon has registered as an external student at a university where he can use the library at any time and have access to the academic community and its publications on-site via PCs and virtual catalogs. This is the ideal complement to the IU online search portal. Today, he is mainly looking for basic literature and high-quality systematic reviews that address his research question. In his work, Simon wants to examine why potential customers of a newly-introduced product had no interest in purchasing it, causing the product to flop. His research questions are as follows: (1) Which factors have a positive as well as a negative effect on the purchasing decisions of young target groups of? (2) How can these findings be applied to the marketing campaign of a product relaunch? To begin with, Simon develops a search strategy and works on a keyword search in his primary language as well as in English in order to find international literature in the databases. Before turning to online databases, he first finds three particularly up-to-date and suitable books in the library catalog. In the course of his research in the online databases it quickly became clear to him that it is important to carefully sort and archive all relevant literature in order to be able to efficiently access it later. Therefore, he downloads a literature management program to his computer, where his sources can be stored and accessed with keywords and other aids. Whenever a source is placed in the text, the program helps to format it in the required citation style. Simon gladly takes the time to properly enter the literature into the software because he knows how important accuracy is so that his own bachelor thesis remains free of errors and plagiarism, i.e., the unfair use of other people's intellectual property.

The handling of literature and sources is a special focus of academic writing. The following contents on plagiarism prevention, database search, search strategies and techniques, literature management, citation and writing guidelines, as well as the bibliography, apply to all IU-specific examination formats. This means that they apply to written assignments and research essays, project reports, case studies, portfolios, bachelor theses, colloquia, and presentations, if applicable. When writing papers at IU, always consult the updated guidelines of the respective courses.

4.1 Plagiarism Prevention

Anyone starting to write an academic paper needs to have comprehensive literature on the subject and adequate access to the internet and academic databases. In today's knowledge-based service society, this is hardly a problem. However, students are faced with the challenge of correctly selecting, processing, qualitatively assessing, systematizing, and evaluating the many approaches to finding relevant literature sources. In many of these freely accessible academic texts, however, authorship is often not clearly recognizable. All the greater is the danger of not respecting the intellectual property of others, since the limits of “yours” and “mine” are less respected due to the low threshold of access (Saunders et al., 2019, p. 115). A university, therefore, has the responsibility to counteract this and develop measures to maintain academic integrity. For this reason, it is important to implement the rules of good academic writing from the start. Plagiarism problems can be avoided. The first part of this unit focuses on raising awareness of this problem in order to prevent any plagiarism from happening in the long run.

What is Plagiarism?

The University of Oxford (2022) defines plagiarism as, “the copying or paraphrasing of other people's work or ideas into your own work without full acknowledgment. All published and unpublished material, whether in manuscript, printed or electronic form, is covered by this definition. 'Collusion' is another form of plagiarism involving the unauthorised collaboration of students or other individuals in a piece of work”.

In the case of plagiarism, a distinction must be made between the various types of plagiarism, e.g., adopting text passages from other authors and others’ ideas, adopting one's own already published ideas (University of Oxford, 2022; Foltýnek et al., 2020, p. 15).

According to Foltýnek et al. (2020, p. 15) plagiarism typically includes:

- the use of someone else’s findings and their presentation as the author’s own findings,
- translation or paraphrase of another work and its presentation as an original work,
- undeclared use of own, formerly published work (self-plagiarism),
- incorrect citations and source referencing,
- undeclared contributions to the work presented,
- undeclared authorship of another person (contract cheating).

Measures and Consequences

The increased digital availability of academic texts brings with it some dangers, e.g., unidentified authorships and the associated increase in plagiarism. In contrast, however, there are also certain advantages that arise from this change because technology significantly increases the possibility of detecting plagiarism. Therefore, many universities now rely on plagiarism software and possible sanctions. IU currently uses the plagiarism detection software **Turnitin** to check all submitted work. Only after this check the examiner is abl to make a final assessment. If the rules of academic work are violated in the form of a detected plagiarism, under the General Examination Regulations (APO) this is treated as an attempt to deceive. In this case, the examination performance is assessed as “unsatisfactory” and is failed.

Turnitin
This is the plagiarism detection software used by IU.

4.2 Database Search

The search for information is a big part of the university degree program. Many sources of information are easily found on the internet, while other sources require a more elaborate search which should be carried out as systematically as possible (see Bramer et al., 2018, for an introduction to this topic). Depending on the resource, there are different ways to access different sources. Naturally, the quality of the results found also varies.

For a first search for identifying relevant literature, the resources of the university library with its range of catalogs and databases should be used (Oliver, 2012, p. 53). So-called discovery systems make it possible to search in different sources s at the same time.

While general online search is usually possible without access restrictions, the use of databases and library catalogs is sometimes reserved for a limited number of users or is only possible for a fee. As a rule, educational and research institutions hold licenses for relevant databases so that authorized persons can use them free of charge (Saunders et al., 2019, pp. 94–95). IU holds licenses for some important databases such as ScienceDirect, WISOnet, EBSCO, and Statista.

Table 5: Important IU Databases/Search Engines

BASE Search Engine	Special search engine of the University of Bielefeld, interdisciplinary
ECONBiz	Research portal for economics, including filter function for full texts, articles from professional, subject-specific journals, working papers (open access), books, and essays
Google Scholar	Special search engine by Google, advanced searches possible and references displayed, i.e., the number of documents that have been quoted in an article: allows conclusions to be drawn about the quality of the article
Karlsruhe Virtual Catalog (KVK)	Metacatalog which simultaneously searches libraries, bookstores, and electronic sources; good to use for an initial overview of the topic
Open-Access.net	Freely accessible sources organized according to subject, e.g. economics, business, psychology

Source: Created on behalf of IU (2020).

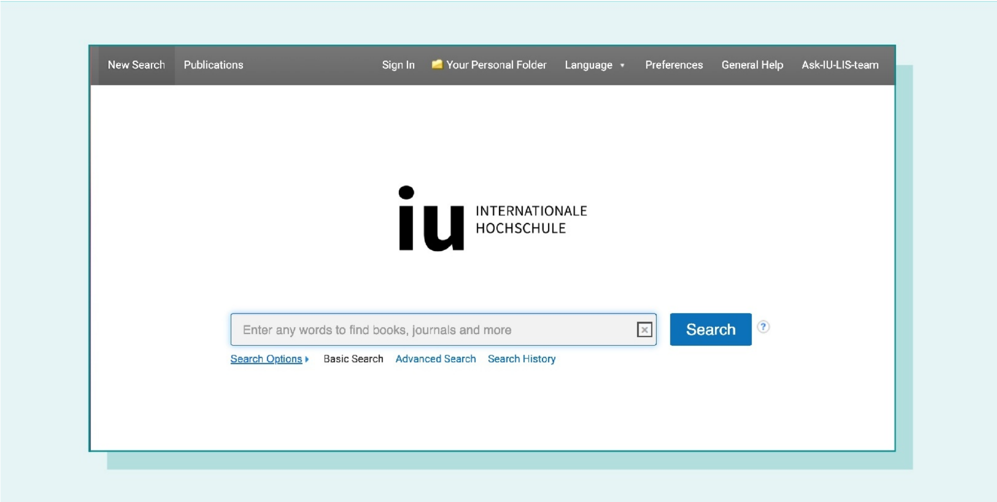
Library catalogs normally contain books and periodicals that are available. Some journal articles are also included. This type of entry can be extensive, for example, with assignment of content-describing keywords, sometimes tables of contents are included. Contents of databases can be quite multifaceted—sometimes you can search full texts of journal articles, press contents, e-books, statistics, company information, and personal information, while other times you only get references to the publications. In most cases, thematic distinctions are present in the databases.

Search Technique

Databases (and general search engines) require their own search language. The simplest versions should be intuitive for all users (Bramer et al., 2018, p. 536).

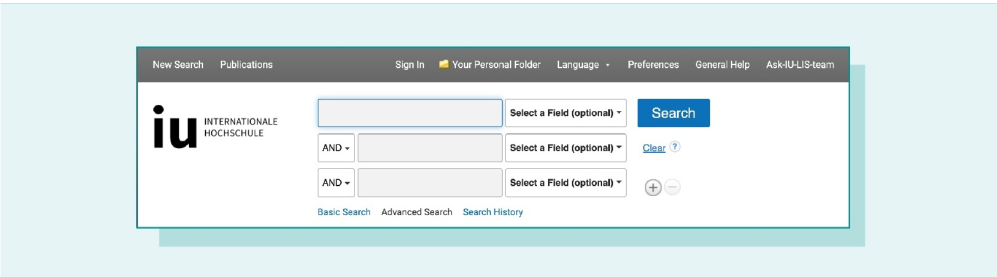
A search term is entered into a search field and then the search is carried out by confirmation ("Search", "OK", or a search symbol). In addition to this Basic Search, Advanced Search also exists in many systems. Here, it is possible to combine search terms and perhaps search in special search fields.

Figure 4: Basic Search Engine Fields, EBSCO Discovery Service IU



Source: Created on behalf of IU (2020).

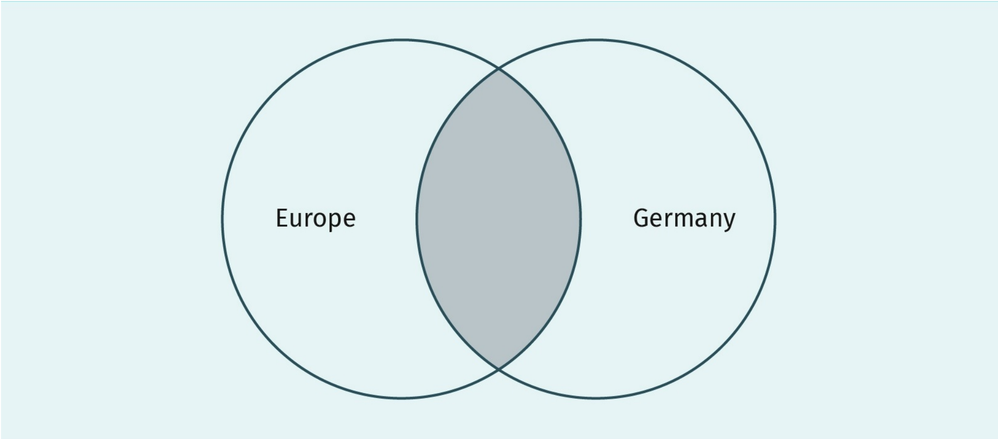
Figure 5: Advanced Search Engine Fields, EBSCO Discovery Service IU



Source: Created on behalf of IU (2020).

In many databases, operators can be used for the search; for example, the Boolean search operators AND, OR, and NOT (Bramer et al., 2018, p. 536). The following illustrations explain the respective effects (the area that is captured is shaded).

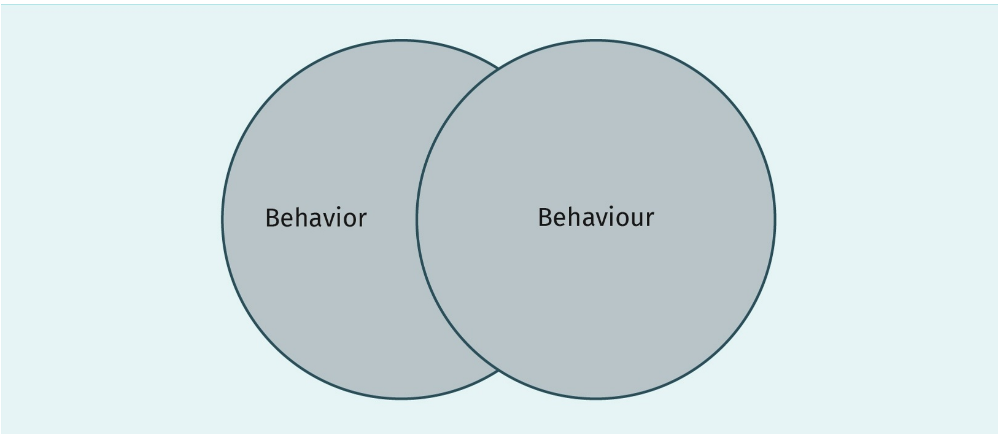
Figure 6: Search Operator AND



Source: Created on behalf of IU (2020).

The result of the search with Europe AND Germany contains the intersection of the search terms. Both search terms appear in every result.

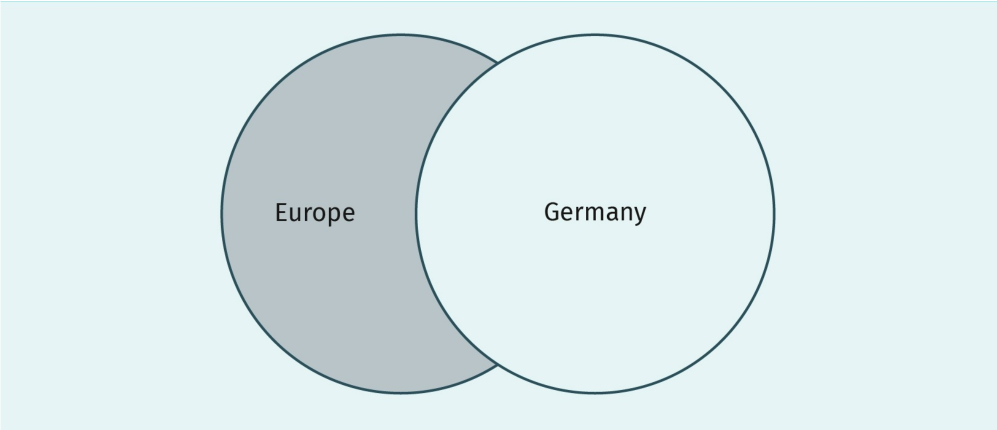
Figure 7: Search Operator OR



Source: Created on behalf of IU (2020).

The result of the search for Behavior OR Behaviour contains the total number of these search terms. Both of the search terms appear in the list of results. This operator is useful for synonyms and terms with different spellings.

Figure 8: Search Operator NOT



Source: Created on behalf of IU (2020).

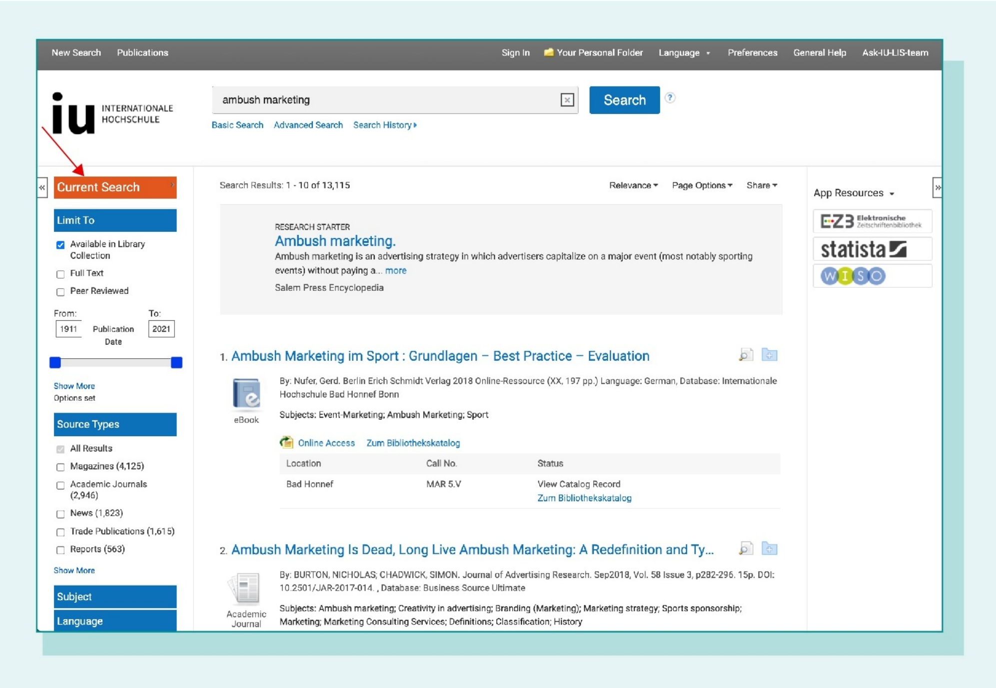
The result of Europe NOT Germany is an exclusion search. The first term should appear in the results list, the second term should not. This shortens the list.

Truncations can replace individual characters or parts of words (Bramer et. al, 2018, pp. 534—535), for example, a search with “europ*” results in hits that contain the search term *Europa*, *Europe*, or also European. The * replaces any character set. A special form of truncation is called masking. Here a wildcard replaces only one character within a search term, e.g., “Ma?er” finds Maier and also Mayer. The characters used for truncation can be found in the help sections of the databases.

For many databases and search engines, the use of “” (quotation marks) is also regarded as activation of the phrase search. With the phrase search, you can ideally search for specific titles or quotations, since the search terms should occur in the order given and with the exact spacing specified.

Discovery systems also have another way to optimize results: lists can be reduced using the filter or faceting options. This can include content (“subject”) or formal criteria (year of publication, media type, etc.).

Figure 9: Filter and Faceting Options (left column) in IU EBSCO Discovery Service



Source: Created on behalf of IU (2020).

Keep in mind that not all search operators are used equally in all search engines or databases. In addition, there are sometimes other operators or search options. Detailed information about which search operators can be used can be found in the help function or FAQ list of the application.

Search Strategies

The process of literature search can be divided in four main steps (Ecker & Skelly, 2010):

1. First step: Formulate an answerable question.
2. Second step: Choose a database, e.g. the Online-Library of Library and Information Services (LIS) or maybe other databases related to the field of study or interest.
3. Third step: Type a word or phrase into the query box.
4. Fourth step: Refine the search.

According to Ecker and Skelly (2010) there are several techniques for refining the search:

1. Replacing general terms with more specific terms
2. Adding terms or combining terms with Boolean operators
3. Using truncate terms or wildcard
4. Use “limit”-Options in the database (e.g., publishing year, language or publication type)

If there is the case, that only a small number of hits is displayed or the literature is not appropriate to answer the question, than it is useful to generalize the search terms or use other search terms.

Another useful strategy to find literature is the snowball search (Florida Atlantic University, 2022). This technique can be used, if one or more relevant sources (e.g., article or book) are already known. This source is used as the starting point to search for additional relevant literature. The search results in literature that listed this source in its bibliography (forward search or citing articles) or the entries from the reference list of this source (backward search or cited articles). In addition, this search strategy allows for a search with keywords from the relevant literature source.

A search strategy that combines different approaches is optimal. It is also helpful to document the performed search steps including the search terms used.

Finding Search Terms

The result of a search, and thus the academic quality of the literature selection, is essentially determined by the search terms used. There are various techniques that are helpfull to find search terms:

- Turning the topic into keywords assist in finding search terms (Walden University, 2021).
- Using synonyms, antonyms and abbreviations for a term/word (Walden University, 2021). Also taking into account different ways of writing or differences between British and American English (organisation vs. organization or colour vs. color) (Oulun Yliopisto, 2022).
- Thesauri are controlled technical vocabulary created by experts that gives inspiration to synonyms, related terms, narrower and/or broader terms (Bramer et al., 2018, pp. 533–534). Thesauri are freely available on the internet, but some are also integrated into databases.
- Keywords can provide ideas for further search terms if relevant literature has already been found (Florida Atlantic University, n.d.).

As a final recommendation for the literature search, it should be mentioned that in search engines or databases there is usually always “something” to find, even if it requires a certain level of skill and amount of practice to determine the sources. It can be helpful to consult with your academic supervisors about what standard research exists in order to get a substantive introduction to your respective topic before proceeding. It is also important to try out the individual resources, to continually improve the search strategy, to refine it with other search terms, and, if necessary, to get help from the database providers or the Library and Information Services (LIS) Team at IU.

Evaluation of Results

All sources that are to be included in an academic paper should first be scanned for certain quality characteristics. Only academic and thematically-appropriate current literature should be used. The *CRAAP Test* can help verify identified sources (Meriam Library at California State University Chico, 2010):

- Currency: Is the information up-to-date?
- Relevance: Is the information important for my question?
- Authority: Who/what is the source of information?
- Accuracy: Has the information been checked for accuracy?
- Purpose: What was the purpose of the information?

These criteria make it relatively easy to identify completely inappropriate publications. The IU Library also offers helpful introductory sessions on database search.

4.3 Literature Management

During all phases of academic work (e.g., in finding and assigning topics, obtaining and processing information, producing texts, or publishing) one is confronted with literature in different formats: books, chapters from books, journal articles, working papers, case studies, etc. (Saunders et al., 2019, pp. 83–90).

So that all this literature is accessible at any given time (e.g., before exams or when writing a research essay or final thesis) and is both organized and easily accessible, it is best to start thinking about how to manage it from the start and create a storage system that allows adding throughout the writing process (Fenner et al., 2014, p. 126).

Optimal literature management begins by recording relevant literature (references). In addition to the traditional index card system, there are various software solutions (Fenner et al., 2014, pp. 122–136), whereby the format for documentation is usually predefined. Since there are different permitted citation styles, the user can chose from a selection list or specifically create one. In many programs, one can differentiate the references by media type and sort them by topic using the folder function. During the writing phase, the previously-entered references can be accessed at any time. For citations, the corresponding source is selected, which is then automatically integrated into the text in the selected citation format. Depending on the citation style chosen, a bibliography in that style is usually created at the end of the document.

Using the word-processing program MS Word as an example, the process is described here. Please note that depending on the version of Word, menu items and individual steps vary. Under the tab “References” you will find the menu item “Manage Sources” or “Insert Quotation”. Entries into your personal bibliography can be made here.

When writing the actual text—depending on the version—the desired entry is selected from the data pool by selecting the “References” tab and the menu item “Insert Quote” or “Quotes”. The citation style can be adjusted at any time via a selection list, e.g., here in APA citation format:

This is the text I am referencing (McGill, 2025, p. 7).

All works cited in the text can automatically be listed in a bibliography at the end. The “References” tab and the “Bibliography” menu item activate the bibliography in the desired format. It can be updated at any time and is a great help especially for longer texts that have been revised more frequently.

External literature management programs (reference management software) usually offer additional functions (Fenner et al., 2014, pp. 126–127). The possibility of transferring literature sources directly from literature databases into the software is noteworthy, making it no longer necessary to enter the title information manually.

There are many reference management software systems such as Bibsonomy, Citavi, CiteULike, EndNote, Mendeley, Papers, Reference Manager, RefWorks, or Zotero. Some of these programs are free; others have to be licensed for a fee. Some are user-friendly and efficient while others require more training, but these usually have more functions (Ivey & Crum, 2018; *Technische Universität München Universitätsbibliothek* (Technical University of Munich University Library, 2020). Generally, the software can be downloaded and installed on the computer, or used on a web interface when internet access is available. Some programs also offer tools for collaboration, such as when sharing sources in a group project. Most programs work with Firefox, Chrome, and Safari browsers.

At IU, students and employees get a campus license for “Citavi for Windows”. Furthermore, the Library and Information Services also offer trainings and consulting services for the free reference management software Zotero. More information about the software and the Citavi license can be found in the Library and Information Services course.

The “traditional” literature management practice of copy-and-pasting one’s literature list into the respective document is also possible. There is no obligation to use literature management software, but it can make working with academic sources much easier, especially in the case of more extensive projects.

The decision for a suitable external literature management program depends on various criteria, such as cost, technical capabilities (hardware or software compatibility), functions, and user-friendliness. Some questions can easily be answered with software comparisons, while others can only be answered by trying.

Note: A literature management program cannot take the place of evaluating the quality or relevance of a source!

Moreover, the use of a reference management program is no guarantee that all relevant information from a source will always be automatically and correctly transfered. Therefore, it is advisable to also check quotations and bibliographies manually at the end.

4.4 Citation and Writing Guidelines

In general, you must acknowledge the use of the intellectual property of others (words or ideas). In all of your academic papers, you must indicate each source and its location. Readers should be able to check each of your sources and to find the cited text passages.

Within the text, citations are written in an abbreviated form. Citations for quotations and paraphrases include the last name of the author(s), the year of publication and the page number. In some types of sources there is no page number. For an audio book, a time stamp is provided, or for an eBook in EPUB format, a chapter is provided. In indirect quotations that do not refer to a particular spot in a text, only the last name of the author(s) and the year of publication are provided.

You should always provide the name of the author in a citation. If your citation comes from two or more pages (e.g. pp. 24 and 25) provide the first and the last page (e.g. pp. 24–25).

To maintain the flow of the text, you can use footnotes to indicate side notes or explanations that are not immediately relevant to the text. Footnotes are also appropriate for translations of quotations from languages other than English or references to copyright. Footnotes are found at the bottom of the relevant page after the last period. They are indicated with a superscript in the text and are numbered throughout the whole text. Start the first word in a footnote with a capital letter, and end it with a period. You should not have long, complex footnotes (especially mathematical proofs or derivations). If you can't integrate these into the text, include them as an appendix to the text. A dedicated footnote page at the end of your text is not necessary.

Generally, you should avoid citing sources that are themselves cited in a source (secondary sources). Always try to find the original source. If you can't access the original source, however, citing a secondary source is acceptable. Cite the original source, then add "as cited in", then cite the secondary source. Include only the secondary source in the reference list.

Table 1

Citation from a secondary source	Cultural tourists can be divided into two groups: specific cultural tourists and general cultural tourists (Irish Tourist Board, 1988 as cited in Steinecke, 2007, pp. 12–13).

Quotations in APA

You can use word-for-word (direct) quotations when the specific phrasing is important for your text or it is very unique. However, to keep your own voice in your writing, do not string too many quotations together. As a general rule, limit the number of quotations in your text. This ensures that the text is your writing and really "yours".

Short Quotations

Quotations of up to 40 words are incorporated into the text with quotation marks.

Table 2

Quotation of a complete sentence	"There is in fact evidence that changes in social practices, and even changes in temporary states of social orientation, can change the way people perceive and think" (Nisbett, 2003, p. 226).
Quotation at the start of a sentence	"Changes in social practices, and even changes in temporary states of social orientation" influence the way people see and think about the world, according to Nisbett (2003, p. 226).
Quotation in the middle of the sentence	Nisbett (2003) writes that "changes in social practices, and even changes in temporary states of social orientation" have an impact on perception and thinking (p. 226). or Nisbett (2003, p. 226) explains that "changes in social practices, and even changes in temporary states of social orientation" have an impact on perception and thinking.
Quotation split into parts	"Changes in social practices, and even changes in temporary states of social orientation" have a significant impact, argues Nisbett (2003), and "can change the way people perceive and think" (p. 226). or "Changes in social practices, and even changes in temporary states of social orientation" have a significant impact, argues Nisbett (2003, p. 226), and "can change the way people perceive and think".

Long Quotations (Over 40 Words)

If the quotation is longer than 40 words, you should format it as an indented (1.27 cm/0.5 inch) paragraph from the left. There are a few other rules:

- Do not use quotation marks.
- Begin the quotation with a capital letter and finish with a period, then follow with the page number in brackets.

Table 3

Long direct quotation	Flores et al. (2018) described how they addressed potential researcher bias when working with an intersectional community of transgender people of color: Everyone on the research team belonged to a stigmatized group but also held privileged identities. Throughout the research process, we attended to the ways in which our privileged and oppressed identities may have influenced the research process, findings, and presentation of results. (p. 311) or Flores et al. (2018, p. 311) described how they addressed potential researcher bias when working with an intersectional community of transgender people of color: Everyone on the research team belonged to a stigmatized group but also held privileged identities. Throughout the research process, we attended to the ways in which our privileged and oppressed identities may have influenced the research process, findings, and presentation of results. or Flores et al. (2018) Everyone on the research team belonged to a stigmatized group but also held privileged identities. Throughout the research process, we attended to the ways in which our privileged and oppressed identities may have influenced the research process, findings, and presentation of results. (Flores et al., p. 311)

Further Rules

- Quotations should be exactly as written in the source. That means that you should also include any typos or grammatical errors, but you can indicate these with a [*sic*]. The first letter of the quotation can be changed to upper or lower case. The punctuation at the end of the quotation may be adjusted to suit the context.
- A quotation within a quotation is indicated with single quotation marks.
- If you leave out any word(s), indicate this with three periods. This does not apply if the quotation begins or ends in the middle of a sentence.
- Place additional comments in square brackets if needed.
- If the quotation emphasizes certain words with italics or bold, but you chose to leave them out, indicate this directly after the quotation [emphasis removed]. If you add your own emphasis you should also in-dicate this [emphasis added] directly after.
- Quotations in languages other than English are included as they are and translated in a footnote.

Table 4

Quotation with a grammatical error	Nowak (2019) wrote that “people have an obligation to care for there [*sic*] pets” (p. 52).
Quotation with emphasis in original	“The *means* being used are ‘co-laboratory’, and academics who receive funding define their research projects for themselves” (Halvorsen & Skare Orgeret, 2019, p. 1).
Quotation with emphasis removed	“The means [emphasis removed] being used are ‘co-laboratory’, and academics who receive funding define their research projects for themselves” (Halvorsen & Skare Orgeret, 2019, p. 1).
Quotation with additional comments	“However, the diverse demographics of tourists and the diversity of their motivations for travel – leisure, recreation, holidays, business meeting, conferences, scientific study, pilgrimage, health treatment – account for the complexity of the phenomenon [of tourism and tourists]” (Goodwin, 2016, p. 6).

Quotation with several words left out	"However, the diverse demographics of tourists and the diversity of their motivations for travel … account for the complexity of the phenomenon"(Goodwin, 2016, p. 6).
Quotation with several words left out at beginning or at end of sentence	"The diversity of their motivations for travel – leisure, recreation, holidays, business meeting, conferences, scientific study, pilgrimage, health treatment – account[s] for the complexity of the phenomenon" (Goodwin, 2016, p. 6).
Quotation with added emphasis	"However, the *diverse demographics* [emphasis added] of tourists and the *diversity of their motivations for travel* [emphasis added] – leisure, recreation, holidays, business meeting, conferences, scientific study, pilgrimage, health treatment – account for the complexity of the phenomenon" (Goodwin, 2016, p. 6).

- Quotations from an audio book should be cited with a time stamp rather than a page number. The time stamp is the point at which the quotation begins.

Table 5

Quotation from an audio book	"There were no visible houses, and I could tell by the state of the road that traffic was very light hereabouts"(Lovecraft, 2011, 5:09).

- Quotations from an eBook in EPUB format that doesn't have stable page numbers are indicated with the chapter (abbreviated as chap.). If the chapter is very long you can add the number of the paragraph (abbreviated as para.).

Table 6

Quotation from an eBook in EPUB format	"The primary objective of developing an autonomous vehicle is to reduce the number of accidents caused by humans" (Ng, 2021, chap. 1.1).

Paraphrases in APA

Paraphrases are also known as indirect quotations, in which you take ideas from one or several sources and put them in your own words. Paraphrases do not use quotation marks. A citation should also be provided for facts or figures that do not come from the literature, but are instead, for example, from your own research. The degree to which this paraphrase relates to your text should be clear.

Table 7

Paraphrase	Sustainably aware consumers in Germany prefer to take holidays in their own country (Klein, 2014, p. 261).

In longer paraphrases that extend over several sentences, the citation should be placed in the first sentence. If it is obvious that the paraphrase also continues in the following sentences, you do not have to include the citation again. This should improve the readability of the text. Care must be taken to ensure that it is obvious from the wording that the same source is used throughout.

However, if the same source is used again in a new paragraph, you must provide the citation again.

Table 8

Longer paraphrase over two paragraphs	In regards to leisure travel, sustainably aware consumers in Germany are characterised by a higher preference for holidays in their own country (Klein, 2014, pp. 261). They also book sustainable travel options more often, and do more research on sustainable tourism. In general it can be said that the biggest differences in the importance of sustainability are seen in the selection of a hotel (Klein, 2014, p. 298).

Multiple Authors

If there are two authors, connect their names with an "and" in the text, or "&" if you are using brackets. If there are three or more authors, then provide just the name of the first author and continue with "et al.". If there are several sources with the same two first authors, you should provide as many names as are necessary to differentiate the sources. More than four authors are just indicated with "et al.".

Table 9

Two authors	
Authors in the text	According to Homburg and Krohmer (2011, p. 50) motivation and needs are closely related. or According to Homburg and Krohmer (2011) motivation and needs are closely related (p. 50).
Authors in brackets	Motivation and needs are closely related (Homburg & Krohmer, 2011, p. 50).
Three authors or more	
Authors in the text	Meffert et al. (2008, p. 98) divide the marketing research process into four phases. or Meffert et al. (2008) divide the marketing research process into four phases (p. 98).
Authors in brackets	The marketing research process can be divided into four phases (Meffert et al., 2008, p. 98).

Two or more sources with the same two first authors with the same publication year	Complete list of authors in reference list: 1. Kapoor, A., Bloom, B., Montez, C., Warner, D., & Hill, E. (2017) 2. Kapoor, A., Bloom, B., Zucker, C., Tang, D., Köroğlu, E., L´Enfant, F., Kim, G., & Daly, H. (2017) In-text citation: 1. (Kapoor, Bloom, Montez et al., 2017, p. 13) 2. (Kapoor, Bloom, Zucker et al., 2017, p. 22)

Citations of Several Sources

If you are referring to several sources (for example, to support an argument), list them alphabetically and separate them with a semicolon. If you are referring to two or more sources from the same author, arrange them chronologically and separate the years with a comma. List the sources with unknown publication dates first.

Table 10

Referring to several sources	A growing number of studies examine the emotions of instructors while teaching (Frenzel, n.d., 2011, 2014; Frenzel et al., 2009; Hagenauer et al., 2015; Keller, 2014, 2016).

Authors with the Same Last Name

To prevent confusion, include the initials of authors that share a last name.

Table 11

Authors with the same last name	The study conducted by A. Klein (2014) found that …

Citing Personal Communication

Include personal (expert) interviews, notes from conversations, letters, emails, internal company documents not available to the public and other personal communication only in the text and **not** in the reference list. If you are quoting a person, ensure that they consent to this first.

Table 12

Citation from an interview	Tölzer Land's tourism organisation sees the diversification of the existing winter offerings as necessary (A. Schmidt, director of Tölzer Land Tourism, personal interview on 20.02.2018, see Appendix XYZ).
Citation from an email	The tourism businesses in the Berchtesgadener Land are mostly small and mid-sized family businesses (P. Müller, Director of Berchtesgadener Land Tourism, email of 20.02.2018).

Citation from an internal report, which is not available to the public	The sales of the hotel’s own restaurant dropped by 70 percent in 2020 compared to 2019 (Hotel zur Sonne, internal financial report from 15/01/2021).

If you have conducted expert interviews you **must** include a transcript of the questions and answers in an appendix, and indicate this in the text (see Appendix….). Each interview must be completely transcribed word for word—do not summarize or write phonetically. However, if there is a section that is out of context, you can use bullet points. Include the name of the person you interviewed, the date and the location (or if appropriate, “by telephone”). Your participants must consent to their names being used. If they do not consent, you can anonymize their names (“Expert A”).

Lecture notes, lecture slides, webinars, etc. are not to be used as sources in academic work.

Organization as Author

If you cannot find the name of the author, provide the name of the organization instead.

Table 13

Organization as author	The Lufthansa Group is divided into the following business segments: passenger airlines, logistics, MRO, catering and other activities (Lufthansa Group, 2018).

If the name of the organization has an acronym, you can include it in brackets in the first citation. In later citations you can just use the acronym.

Table 14

First citation	In 2017 Germany held the ninth position in international arrivals numbers (United Nations World Tourism Organization [UNWTO], 2017).
Further citations	For several years France has held the first position (UNWTO, 2017).

If you cannot identify the name of the organization, use the title instead.

Table 15

Unknown organization	“List of Oldest Companies”(2019) or (“List of Oldest Companies”, 2019)

Unknown Date

If you cannot identify the year of publication, use “n.d.” (no date).

Table 16

Unknown publication date	Ray und Anderson (n.d.) estimated the proportion of sustainably aware consumers in the USA in 2008 to be about a third.

Online Sources

Online sources are cited exactly like books or journals: in the text you give an abbreviated citation and in the reference list the full one. The title or web address is not included in the citation. A general reference to a website does not need a citation or an entry in the reference list. Instead, provide the name of the site in brackets in the text.

Table 17

Citation from an online source	Following the fusion with the Adam Ries University of Applied Sciences in September 2013, the study programmes were offered under the IUBH Dual Studies Munich name (IUBH Internationale Hochschule GmbH, 2018).
General reference to a webpage	The survey was created using the Unipark (www.unipark.de) online survey tool.

4.5 Bibliography

Any academic work must include a reference list at the end. This is the complete list of all the academic sources that you mentioned in your work as well as other materials that you have referred to.

The reference list provides a transparent list of the sources you used for the text. Your reader must be able to locate and check each source. In order to do so, the following rules apply: accuracy (the source must be correct); completeness (provide enough information that makes it possible to find the source); consistency (use a particular format throughout); and clarity. List sources in **alphabetical** order by author, and chronologically by publication date in the case of several sources from one author. After the last name of the author, provide their initials and the year of publication in brackets. Space the list at 1.5 and align with the left margin; from the second line a reference is indented by 1.27 cm. Beyond these rules the following also apply:

- Do not use dashes, hyphens or bullet points.
- The names of books, journals or newspapers are in italics.
- Titles of all works, such as books, articles and webpages are written in sentence case in the reference list, even if title case has been used in the original work. Use “:” to separate title and subtitle.
- If one author has several publications in one year, add a, b, c … after the year of publication.
- If an author appears several times in your reference list, work in chronological order.
- If you do not have a publication date, indicate this with “n.d.” (no date).
- If there is no information about the volume of a journal, leave this out altogether.

- If you do not have the name of the author, provide the title instead, then follow with the year or the date and the rest of the reference.
  - Example **with** author: Sapolsky, R. M. (2017). *Behave: The biology of humans at our best and worst*. Penguin Books.
  - Example **without** author: *Behave: The biology of humans at our best and worst.* (2017). Penguin Books.
- Do not separate sources by types such as print, online, etc. All sources are in one list.
- Separate authors by a comma. From two to 20 authors include "&" before the last author.
- From 21 authors or more provide the first 19, then type "…", then the name of the last author.
- If a source is the first edition, you do not need to indicate the edition.
- Do not include academic titles of the author(s) or editor(s).
- In the case of organizations as authors or editors, a period should be placed after the name of the organization (Institution. (year))

Particular Rules for Electronic Online Sources

- If possible, provide the digital object identifier (DOI). If there is no DOI, provide the URL. If you're using an electronic version of a source that is also available in print, cite the electronic version.
- The URL should be a permalink that leads the reader directly to the source. As this is not always possible, in certain cases the procedure is different:
  - If you accessed a source through a password-protected online database or eBook platform and it does not have a DOI, do not provide the URL. Cite as you would for a print version.
  - If you use a one-time link to an online session (e.g., collecting statistical data on Destatis) provide the URL as precisely as possible. If this is not an option, use the login or main page. You should explain in the text which data you used.

Monographs in APA

eBooks are cited the same way as print books. Include the DOI of an electronic version of a source if it is available. If there is no DOI for a freely available source, provide the URL. If the source is not freely available online and there is no DOI, provide the same information as you would for a printed source.

One author: Last name, initials. (Year). *Title: Subtitle* (Edition [if later than the first edition]). Publisher. DOI (if available)

Two authors: Last name, initials., & last name, initials. (Year). *Title: Subtitle* (Edition [if later than the first edition]). Publisher. DOI (if available)

Three to twenty authors: Last name, initials., Last name, initials., & Last name, initials. (Year). *Title: Subtitle* (Edition [if later than the first edition]). DOI (if available)

21 authors or more: Last name, initials of the first author., Last name, initials of the second through nineteenth author., … Last name, initials of the last author. (Year). *Title. Subtitle* (Edition [if later than the first edition]). DOI (if available)

Table 18

One author	Fletcher, D. P. (2018). *Disrupters: Success strategies for women who break the mold*. Entrepreneur Press.
Two authors	Nicol, A. A. M., & Pexman, P. M. (1999). *Presenting your findings: A practical guide for creatingtables.* American Psychological Association.
Three to twenty authors (electronic version with DOI)	Taha, W. M., Taha, A.-E. M., & Thunberg, J. (2021). *Cyber-Physical Systems: A Model-BasedApproach*. Springer Nature. https://doi-org.pxz.iubh.de:8443/10.1007/978-3-030-36071-9
21 authors or more	Gabel, H., Müller, J., Ilsemann, U., Georgen, K., Kaffenberger, N., Lagemann, E., Meyer, K., Macke, D., Schmidbauer, S., Paffel, S., Jürgens, T., Tannenberg, F., Dannenberg, R., Raabe, T., Corvinus, A., Hofbauer, W., Becker, A., Schumacher, C., Radanovich, A., … Schmitz, L. (2001). *Foundations of Sustainable Development* (2nd ed.). Econ Publishing.

Chapters in Edited Books

eBooks are cited in the same way as printed books. Include the DOI of an electronic version of a source if it is available. If there is no DOI for a freely available source, provide the URL. If the source is not freely available and there is no DOI, provide the same information as you would for a printed source.

Indicate the authors in the same way as explained for monographs.

One editor: Last name, initials. (Year). Title: Subtitle. In initials. last name (Ed.), *Title: Subtitle* (Edition [if later than the first edition]), pp.?–?). Publisher. DOI (if available)

Two editors: Last name, initials. (Year). Title: Subtitle. In initials. last name & initials. last name (Eds.), *Title: Subtitle* (Edition [if later than the first edition]), pp.?–?). Publisher. DOI (if available)

Three or more editors: Last name, initials. (Year). Title: Subtitle. In initials. last name, initials. last name, & initials. last name (Eds.), *Title: Subtitle* (Edition [if later than the first edition]), pp.?–?). Publisher. DOI (if available)

Table 19

One editor (electronic version with DOI)	Young, R. A. (2019). A Contextual Action Theory of Career. In J. G. Maree (Ed.), *Handbookof Innovative Career Counselling* (pp. 19–33). Springer Gabler Verlag. https://doi.org/10.1007/978-3-030-22799-9_2
Two editors	Rattan, A. (2019). How lay theories (or mindsets) shape the confrontation of prejudice. In R. K. Mallett & M. J. Monteith (Eds.), *Confronting prejudice and discrimination: Thescience of changing minds and behaviors* (pp. 121–140). Academic Press.

Three or more editors (electronic version with DOI)	Aron, L., Botella, M., & Lubart, T. (2019). Culinary arts: Talent and their development. In R. F. Subotnik, P. Olszewski-Kubilius, & F. C. Worrell (Eds.), *The psychology of high-performance: Developing human potential into domain-specific talent* (pp. 345–359). American Psychological Association. https://doi.org/10.1037/0000120-016

Journal and Newspaper Articles in APA

Include the DOI of an electronic version of a source if it's available. If there is no DOI for a freely available source, provide the URL. If the source is not freely available and you got it from a database, provide the same information as you would for a printed source.

Last name, initials (Year). Title: Subtitle. *Name of journal or newspaper, volume* (number or issue), first page of article-last page of article. DOI (if available) or URL

Do not include the abbreviation p. or pp. before page numbers for journal or newspaper articles in the reference list. If the article does not have volume, issue or page numbers (e.g., because it is published on the website of a newspaper or as an "Online First" journal article that is not yet assigned to a specific issue) omit these elements.

Provide complete dates for newspaper articles rather than just the year.

Table 20

Printed version	Sutton, R. I., & Staw, B. M. (1995). What theory is not. *Administrative Science Quarterly, 40*(3), 371–384.
Electronic version, DOI available	Ludwig, J., Duncan, G. J., Gennetian, L. A., Katz, L. F., Kessler, R. C., Kling, J. R., & Sanbonmatsu, L. (2012). Neighborhood effects on the long-term well-being of low-income adults. *Science*, *337*(6101), 1505–1510. https://doi.org/10.1126/science.1224648
Online newspaper article	Carey, B. (2019, March 22). Can we get better at forgetting? *The New York Times.* https://www.nytimes.com/2019/03/22/health/memory-forgetting-psychology.html

Online Sources in APA

Referencing online sources is much the same as referencing printed sources. You should provide the author of the document, the year it was last updated (on a homepage, the last year the copyright was updated), *the title,* the name of the website or the institution and the exact location. You don't need to provide the date you retrieved the information. The location is the DOI, if available; otherwise provide the complete URL (if available, the permalink). The DOI is much like a serial number and so gives digital work a unique and permanent identifier.

If no author is available, provide the name of the institution that publishes the work instead. In this case leave the name of the website out if it is identical to the name of the institution. In your text, online sources are cited in the same way as print sources. If you cite several pages of a website, you should have a separate reference list entry for each.

Follow these rules to refer to online sources in the reference list:

- If you can identify the author: provide their name
- If you can identify the name of the institution: provide it (you can find this in the imprint page, or terms of service or legal sections)
- In other cases, provide the name of the website.

Table 21

PDF document (with report number)	Stuster, J., Adolf, J., Byrne, V., & Greene, M. (2018). *Human exploration of Mars: Preliminarylist of crew tasks* (Report No. NASA/CR-2018-220043). National Aeronautics and Space Administration. https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20190001401.pdf
PDF without an author (government agency)	Federal Trade Commission (2017). *Privacy & Data Security: Update: 2016*. https://www.ftc.gov/system/files/documents/reports/privacy-data-security-update 2016/privacy_and_data_security_update_2016_web.pdf
Webpage	Mischel, W. (n.d.). *Psychology*. Encyclopedia Britannica. https://www.britannica.com/science/psychology
Several pages from one website (referencing several items from the same year)	United Nations. (2021a). *Deliver Humanitarian Aid.* https://www.un.org/en/our-work/deliver-hu-manitarian-aid United Nations. (2021b). *Maintain International Peace and Security.* https://www.un.org/en/our-work/maintain-international-peace-and-security
Statistics from Statista	Centers for Disease Control and Prevention (CDC) (2021). *Distribution of the 10 leadingcauses of death among American Indians or Alaska Natives in the United States in 2018*.* Statista. https://www-statista-com.pxz.iubh.de:8443/statistics/233320/distribution-of-the-10-leading-causes-of-death-among-american-indians-or-native-alaskans/
Preprint article from a repository/publication server	Baumli, K., Warde-Farley, D., Hansen, S., & Mnih, V. (2020). *Relative Variational IntrinsicControl.* arXiv. https://arxiv.org/pdf/2012.07827.pdf

Multimedia Sources in APA

Podcast episode: Name, initials. (Role—host, producer, author, etc.). (year, month day). Title: Subtitle (episode number, if available) [Type of podcast episode: video podcast episode, audio podcast episode). In *Name of podcast series* (if available). Production company. URL

Motion picture: Name, initials. (Role—director, screenwriter, etc.). (Year). *Title: Subtitle* [Motion picture]. Studio/production company (if several, separate with semicolons).

Video on an online platform: Name of the account. (year, month day). *Title of video* [Video]. Platform. URL

Dates are the date of release or the most recent update. If there is no date, type n.d.

An audiobook is referenced in the same way as a print book, so you do not need to indicate that it is an electronic version. This is not the case if:

- The content of the audiobook is different from the print version (for example, an abridged version).
- You have a reason to indicate that it is an audiobook (for example, if you are studying the impact of the narration on the listener).
- You have cited the audiobook in your text with a timestamp (see section 2.1.1).

Audiobook = Print book: Name, initials. (year). *Title: Subtitle*. Publisher.

Audiobook ≠ Print book or audiobook quoted in text: Name, initials. (year). *Title: Subtitle* [audiobook]. Publisher.

Table 22

Podcast	Hannah-Jones, N. (Host). (2019, September 13). How the bad blood started (4) [Audio pod-cast episode]. In *1619*. The New York Times. https://podcasts.apple.com/us/podcast/epi-sode-4-how-the-bad-blood-started/id1476928106?i=1000449718223
Video on an online platform	Lushi, K. [Korab Lushi]. (2016, July 3). *Albatross culture 1* [Video]. YouTube. https://www.youtube.com/watch?v=_AMrJRQDPjk&t=148s
Motion picture	Ross, G. (Director & screenwriter), & Collins, S. (Screenwriter). (2012). *The hunger-games* [Motion picture]. Lionsgate.
Audiobook ≠ Print book	Cain, S. (2012). *Quiet: The power of introverts in a world that can't stop talking* (K. Mazur, Narr.) [Audiobook]. Random House Audio.

All rules for correct citation and referencing can be found in the General Citation Guidelines of IU.

SUMMARY

In today's increasingly digitalized knowledge-based service society, dealing thoughtfully and professionally with literature references is both challenging and important. This unit reviews the most important expectations regarding efficient searches and systematized literature when writing scholarly papers, with special emphasis placed on the avoidance of plagiarism. The unit explains how to meet these requirements by using literature management software and uploading texts to the plagiarism software Turnitin, as well as which citation rules and writing guidelines must be followed to ensure quality work.

UNIT 5

ACADEMIC WORK AT IU: WRITTEN ASSIGNMENT AND RESEARCH ESSAYS

STUDY GOALS

On completion of this unit, you will have learned ...

- how a research essay at IU is structured.
- about different types of instructions to create a written assignment and a research essay.
- which formalities have to be observed when writing a written assignment and a research essay.
- what requirements are put on the scope and content of a written assignment and a research essay.

5. ACADEMIC WORK AT IU: WRITTEN ASSIGNMENT AND RESEARCH ESSAYS

Case Study

Maike has to write a research essay. She wants to explore "Equal economic opportunities in the time of digitalization and technology". This is a topic that Maike is keenly interested in because she herself wants to have a career and find a way to balance that with also having a family. It therefore makes sense to address the opportunities, hurdles, and challenges from an objective perspective rather than from a purely subjective perspective. She collects search terms for an initial search in the online library and develops a common thread—an initial rough outline for the introduction, body, and conclusion of her essay. In order to avoid misunderstandings, Maike once again reviews the differences between a written assignment and a research essay.

5.1 Written Assignments and Research Essays at IU

The written assignment and research essay belong to the various forms of academic work that are written in almost every course of study. In many classes, research essays are required. This has, among other things, the purpose of preparing the student for the final bachelor thesis. Although the contents and structures of these works differ among scientific disciplines because subject-specific traditions play a role, the most important aspects are quite similar and in many cases are also congruent. The following statements therefore represent a "lowest common denominator" and also reflect the formal requirements of IU. The specific requirements can be found in the Exam Guide in myCampus for research essays/written assignments and the specific tasks in the particular courses.

Independent scienific working is learned through writing such assignments and research essays. For this purpose, it is first necessary to choose a scientific topic, to specify a research question, to search for, and evaluate, appropriate literature, and, finally, to present and theoretically substantiate one's own thoughts and findings.

Structure

The structure of written assignments and research essays typically follows a classical pattern:

1. Introduction
    a) Rationale for chosen topic
    b) Aim of the work

    c) Definition of the topic and research question to be addressed
    d) Overview of structure
2. Body
    a) Identify main points
    b) Explain, discuss, and supplement main statement(s) with additional statements
    c) Draw conclusions from the reasoning (which lead to the next step)
3. Conclusion
    a) Summary of argument
    b) Summary with student's own conclusions
    c) To what extent has the aim of the work been achieved?
    d) Statement with further open questions and/or perspectives

This is a general structure but does not mean that each section title—introduction, body, and conclusion—must be chosen. The body should be given a different title. However, it is important to internalize the basic structure of the written assignment/research essay. The structure of the written assignment/research essay is determined by the logical sequence of the main points and explanatory steps. The entire text should be closed and complete in terms of content and should have a straightforward structure. Logical transitions must be created between the individual sections by rephrasing the central statement of the previous section, in order to lead to the objectives of the next section. The core of the work is the body, which is framed by an introduction and a conclusion. The following approximate proportions can be assumed to give a rough orientation frame:

- introduction, approximately 20 percent of the work
- body, approximately 70 percent of the work
- conclusion, approximately 10 percent of the work

This serves as a rough outline; it all very much depends on the specifics of each written assignment and research essay.

Formal Guidelines and Specifications for Submission

In addition to the sections previously described, IU has other **formal components** which must be incorporated into every written assignment and research essay.

Formal components
These are the elements that must be included.

A written assignment and research essay consists of the following parts (in this order):

- title page
- table of contents
- list of figures and/or tables (if necessary)
- list of abbreviations (if necessary)
- text part with introduction, body, conclusion
- bibliography
- list of appendices (if necessary)
- appendices and materials (if necessary)

Formal Document Requirements

The following formal guidelines apply to written assignments and research essays.

Table 6: Formal IU Requirements for Written Assignments and Research Essays

Length	7–10 pages of text
Paper size	DIN-A4
Margins	Top and bottom 2 cm; left 2 cm; right 2 cm
Font	General text: Arial 11 pt.; Headings: 12 pt. justified
Line spacing	1.5
Sentence structure	Justified and use of hyphens to separate words
Footnotes	Arial 10 pt. font, justified
Paragraph	6 pt. distance after line break
Section/subsection levels	Maximum three levels (1. Main heading, 1.1 Section, 1.1.1 Subheadings)
	Only individual chapters in the text are numbered consecutively; otherwise, sections of the assignment, such as the list of figures or the bibliography, are not numbered.
	Do not use the underline function, and use italics sparingly to emphasize passages.

Source: Created on behalf of IU (2020).

The written assignment or research essays should be submitted to the Turnitin online portal. The exact steps of submission can be found in a separate manual on myCampus. There it is also explained how the evaluation can be viewed directly on Turnitin after the publication of the grade on myCampus. It is not possible to deliver the examination by email or other means.

It is important that the affidavit is first submitted electronically via myCampus. Prior to this, it is not possible to submit the work. Further information can be found in the "Instructions for Submissions in myCampus—Turnitin".

Written assignments and research essays differ both in the requirements and in the didactic focus. Written assignments are only demanded in courses with course book, research essays are demanded in courses without course book. The specific requirements are explained below.

Topic and Task of the Written Assignment

For a written assignment, a task is assigned or one can select from several alternative tasks. It is important to thoroughly examine the task and to try to align the work exactly. It is not sufficient for the written assignment to use only the course book as a source of information; rather, it is expected to use additional relevant sources. These can be found, for example, in our online library databases.

In written assignment task, several support channels are open; as the student, it is your responsibility to select your preferred support channel. The tutor is available for technical consultations and for formal and general questions regarding the procedure for processing the written assignment. However, the tutor is not required to approve outlines or parts of texts and drafts. Independent preparation is part of the examination work and is included in the overall evaluation. However, general editing tips and instructions are given in order to help you get started with the written assignment.

Evaluation of the Written Assignment

The assessment of a written assignment contains several components.

Table 7: Evaluation of a Written Assignment

Criterion	Explanation
Introduction	Introduction and selection of topic
Structure	Structure and design
Reasoning	Quality of reasoning and research
Conclusion	Conclusion and recommendations
Language	Linguistic expression and spelling
Tidiness	Cleanliness in presentation and citation

Source: Created on behalf of IU (2020).

Topic and Task of a Research Essay

There are different subject areas to choose from. For the concrete selection of the essay topic, there are two possibilities: Choosing one of the already given questions or coming up with own ideas within the subject area. In this case, it has to be considered that the question must be agreed and approved upon in advance with the tutor. The independent delimitation and formulation of a question represents a good preparation and exercise, especially regarding the later thesis. Introductory literature references are given for each of the subject areas, which can serve as a starting point. It is expected that further literature sources will be researched and incorporated into the research essay. It should be noted that the theoretical approach and the related literature review are the main focus of the research essay.

The tutorial support in research essays is similar to the support in written assignments.

Evaluation of a Research Essay

The demands of a research essay are higher than those of a written assignment, specifically with regard to level of personal contribution and the scientific structure.

Table 8: Evaluation of a Research Essay

Criterion	Explanation
Introduction	Structure of the objectives
Structure	Development and process
Theory	Literature analysis, application and understanding of definitions and precedents, quality of sources
Methodology	Clear information on the chosen methodology, justification, correct application, use of theoretical and practical aspects
Reasoning	Quality of argumentation/conclusive development, clarity and accuracy of reasoning, topic comprehension, connection between theoretical and practical, critical assessment of the findings and own interpretation
Conclusion	Conclusions and recommendations
Presentation	Cleanliness of the presentation
Accuracy	Spelling/punctuation
Language	Linguistic expression
Literature	References/literature list

Source: Created on behalf of IU (2020).

Further specification and guidelines can be found in the respective Exam Guide in myCampus and the guidelines of particular courses.

UNIT 6

ACADEMIC WORK AT IU: PROJECT REPORTS

STUDY GOALS

On completion of this unit, you will have learned …

- how a project report at IU is developed.
- which instructions to use to write a project report.
- which principles need to be observed when writing a project report.
- the requirements encompassing both the scope and content of a project report.

6. ACADEMIC WORK AT IU: PROJECT REPORTS

Case Study

As part of a pilot project at zielNET, a superfood product will be introduced with new marketing. A peer group will accompany the entire relaunch for 12 months. The idea to include the input of a “reflective team” originates from the increasingly popular theory of agile process methods. This means that a group of approximately 15 people will, at regular intervals, provide feedback on the steps for the live and on-site relaunch. The goal is to initiate a permanent improvement process so that the market launch of the new superfood product will be successful. Simon, a student at IU, will be heavily involved in the project as project coordinator, and will be using this opportunity to write a project report, which he will publish in the Marketing module. After an initial brainstorming session with his boss and project manager, Simon asks the following questions: What should the motive for the project report be? What must be included in each case? As far as the motives are concerned, the project report is certainly first and foremost about recording the course of a project as accurately as possible, i.e., mapping the entire process in its individual process steps and associated tasks. In this context, Simon remembers the PDCA cycle (Plan, Do, Check, Act). Projects can be planned, developed, implemented, and reviewed with the help of this management tool. Supplemental files containing a collection of resources, all flow into the project report. Simon quickly realizes that a project report is a valuable document and sees the importance of this exercise, challenging him to combine theory and practice together.

6.1 The IU Project Report

The performance review of a project report is to see that the student is able to successfully develop a project that is independently conceived, implemented, and written. If necessary, this process can be supervised by the respective tutor. The difference from a research essay or written assignment lies in the project part—the practical aspect that precedes the project report.

The project report as a way to test student knowledge combines expertise with its transfer into practice on the basis of a concrete project, such as the transfer to a concrete practical problem in a real workplace.

The result of a project report always consists of a “product” in the actual or figurative sense. This can be of a physical nature (e.g., an engineering model) or can also consist of a concept, software solution, installation, process, or the like.

The project report should provide information as completely, precisely, and comprehensibly as possible about the individual work, development steps, and approach. The project report is thus the protocol of the entire project management process with its intermediate steps and products, including a reflection on the approach and methodology. Part of the project report is also to procure the necessary resources for implementation. These may consist of data, surveys, applications, technical equipment, software, various tools, etc.

The end product shall also be suitably and appropriately documented and, if necessary, the use in a typical user situation simulated and exemplified. Important: The selection, **parameterization**, and use of programs (software, tools, apps, etc.) is also an integral part of the project for conceptual topics.

Parameterization
Guidelines and reference values are types of parameters.

The Structure of a Project Report

In order to develop a product in the sense of the project report, a multi-stage procedure is necessary. These steps are reflected in the project report in various sections and subsections, which can be oriented to the following structure. Depending on the focus and scope of the project, **deviations from this structure** are also possible. It is therefore not their strict adherence that is central, but rather an adequate adaptation to the specific project carried out in each individual case. The orientation to central questions, which serve as a framework during the project, can also serve to aid the development of a suitable structure of the project report. The project report is then created as completely as possible by answering these key questions as a continuous text with (partial) headings.

Deviations from the structure
A fixed structure is not as necessary in a project report as in other written formats.

The definition of the structure in meaningful sections, subsections, and units can differ depending on the project and the part to be addressed. This includes, for example, the naming of project partners or sponsors.

The basic structure typically adheres to the following pattern:

- Introduction: Presentation of the project objectives and preparatory measures as well as the tools used
- Body: Presentation of the procedure for project implementation
- Conclusion: Summary of the results achieved

This does not mean that these headings—Introduction, Body, and Conclusion—should be chosen for the three sections. The body of the text especially should be named differently. However, it is important to understand the basic structure of the project report.

Introduction

In the **introduction** it is first necessary to define the problem, analyze the initial conditions, and outline the objectives. At the beginning of the introduction, the project idea should be briefly presented, with a short description of the content and goal(s) as well as, if needed, a description of the underlying context, project environment, and initial motivation. This part should comprise of 10–20 percent of the report. Part of the introduction is also a brief explanation of the process for solving the problem, e.g., by giving a short overview of the individual chapters.

Introduction
The introduction provides a brief overview, key objectives, and context about the project.

The preliminary planning and the process is also described. The following key questions can serve as guidance for the introductory section:

- What was the goal/what were the goals of my work?
- How did I go about it? What preparatory measures have been taken? (data collection, data analysis, evaluation of material, etc.)
- Which planning/organizational activities were necessary?
- Why did I proceed this way?
- Which methodology/tool did I choose and why?

Body

Body
The body of the project report presents the concrete project in its development, implementation, and review. The results and resources play an important role in this process.

The **body** of the project report should describe the actual implementation of the project. This includes special milestones and challenges that may have arisen during the course of the project. All (development) steps must be recorded in the project report; interim results should be recorded. A further requirement is that the project be carried out efficiently and with the available, limited resources. Experience has shown that the body of the report should comprise around 70–80 percent of the report. Keywords for the description of the project implementation are:

- stakeholders
- risk analysis
- main tasks: phase planning/project phases
- project structure plan
- project progress/progress report
- sequence and scheduling of important dates and events (start/end event, start/end date)
- project resources: resource/cost planning

Conclusion

Conclusion
This part entails a reflection and a critical analysis of the whole project.

The **conclusion** of the project report is dedicated to the project results, i.e., the evaluation. It should comprise about 10–20 percent of the report and contain both a detailed reflection and critical analysis of the project when evaluating its results. The following points should be incorporated when crafting the conclusion:

- Achievement of objectives: What result was achieved? Does the result correspond to the goal of the work? If not, why not?
- What conclusions can be drawn from the course of the project? What can be improved, if needs be?
- Where appropriate, provide reflections on efforts made and project due dates.
- Where appropriate, provide reflections on available resources and costs.

Formal Guidelines and Submission Requirements

In addition to the written portion, there are other formal elements to be included in every project report. A project report consists of the following parts (in this order):

- title page
- table of contents
- list of figures and/or tables (if necessary)
- list of abbreviations (if necessary)
- text part with introduction, body, conclusion
- bibliography
- list of appendices (if necessary)
- appendices and materials (if necessary)

Formal Document Requirements

The following formal requirements apply to the text of the project report:

Table 9: Formal IU Project Report Requirements

Length	7-10 pages of text
Paper size	DIN-A4
Margins	Top and bottom 2 cm; left 2 cm; right 2 cm
Font	General text: Arial 11 pt., Headings: 12 pt., justified
Line spacing	1.5
Sentence structure	Justified, with auto-hyphenation
Footnotes	Arial 10 pt., justified
Paragraphs	6 pt. distance after line break
Section/Sub-sections	Maximum three levels (1. Main heading, 1.1 Section, 1.1.1 Sub-headings)
	Only individual chapters in the text are numbered consecutively; otherwise, sections of the assignment, such as the list of figures or the bibliography, are not numbered.
	Do not underline; use italics sparingly to emphasize passages

Source: Created on behalf of IU (2020).

The submission of the project report happens via the Turnitin portal. Instructions on how to submit this can be found in a separate manual on myCampus. There you can also find out how the evaluation can be viewed directly on Turnitin after the publication of the grade on myCampus. It is not possible to deliver the information by email or by other means.

It is important that the affidavit first be submitted electronically via myCampus. Prior to this, it is not possible to submit your report. Further information on this can be found in the “Instructions for Submission in myCampus—Turnitin”.

Task

For the project report several topics are proposed. Typically, students are expected to describe a theory-based, conceptually-structured implementation of the project, ideally linking it to the student's personal work context as much as possible. Details on the topic and tasks of the project reports are then given in the respective courses. Any questions regarding the task can be discussed with the responsible tutor.

In this project report task, several support channels are open; as the student, it is your responsibility to select your preferred support channel. The tutor is available for technical consultations and for formal and general questions regarding the procedure for processing the project report. However, the tutor is not required to approve outlines or parts of texts and drafts. Independent preparation is part of the examination work and is included in the overall evaluation. However, general editing tips and instructions are given in order to help getting started with the project report.

Further specification and guidelines can be found in the respective Exam Guide in myCampus and the guidelines of particular courses.

UNIT 7

ACADEMIC WORK AT IU: CASE STUDIES

STUDY GOALS

On completion of this unit, you will have learned …

- how a case study at IU is structured.
- what instructions to use to create a case study.
- what formalities need to be observed when creating a case study.
- what requirements encompass both the scope and content of a case study.

7. ACADEMIC WORK AT IU: CASE STUDIES

Case Study

Maike becomes a member of the Business Ladies—a network for young, female executives that operates internationally. During the first meeting, she meets Tilda, a 38-year-old mother of a 13-month-old son who was, until 13 months ago, the head of the public relations department at an energy company with approximately 1,000 employees. With the birth of her son, however, she was immediately removed from her managerial position by the company's management, which is of the opinion that women—just like men—can only exercise leadership positions on a full-time basis. Tilda agreed to a 75 percent part-time position at the end of her maternity leave. After eight years in a company-wide management position, she now is only in charge of the administration in her own department. Maike and Tilda agree that this is unfair and possibly illegal, but should Tilda risk her current job for a lawsuit now that she is more dependent than ever on the income it provides? Maike would like to write a case study about Tilda's situation, a requirement for her studies in the Management Techniques module. She asks Tilda if she would be willing to participate in a more in depth interview. The "Tilda" case is to be portrayed systematically. With the help of the interview and previously developed key questions, the identified shortcomings, needs for action, and draft political laws for the future will be discussed and reflected upon.

7.1 The IU Case Study

Case studies have a prominent place in academics. What is the value of a case study for students? The value lies, in particular, in the fact that a concrete practical case serves as a basis for systematically applying the theories conveyed in the course, i.e., to analyze the case with that information and, if possible, to find a solution. Case studies are often complex and do not provide all the details the solution would require. This simulates situations in professional practice which, especially in the case of major challenges, only limited contextual information is available. Managers must be able to deal with missing information—therein lies the value in case studies.

The analysis and processing of case studies plays an important role in modern studies because they promote important competences in solution-oriented thinking and decision-making. Case studies also train the practical transfer of theoretical knowledge and models learned in (self-)study. Case studies are a teaching method in which students work on a specific "case" that represents a practical problem situation.

Case studies occur in a wide variety of professional contexts and are used for various purposes in professional practice, studies, teaching, and research. Classical case studies are often reflections on actual decisions from the past based on real data. However, fictitious company and organizational cases are equally suitable for this purpose.

Case studies do not contain a structured preparation of knowledge as textbooks do. The task of case studies can be seen more like real life: complex, incomplete, unstructured, imprecise, and ambiguous.

General Learning Objectives

Case studies seek to gain useful insights through the analytical consideration of example questions from the respective professional practice. Ideally, these findings can be abstracted and transferred to other cases and situations. In this way, case studies support the development of analytical skills and sharpen the ability to separate the important from the unimportant and to open up new alternatives for action.

Structure and General Instructions

A case study should follow the classic pattern:

1. Introduction (case context and explanation)
2. Body (case presentation, processing, and solution)
3. Conclusion (discussion and further transfers)

Of course, this does not mean that these headings—Introduction, Body, and Conclusion—should be the only ones chosen for the three sections; other heading titles are acceptable as well. However, it is important that the case study has a logical structure that is clearly comprehensible to the reader. Further suggestions on how to structure the case study are not possible here due to the variety of possible case studies to use in one's coursework. Case studies have special features that can also lead to challenges while working on them. Such challenges are desired because case studies exist to prepare students for situations that can occur in everyday professional life. This also includes the fact that case studies often do not contain all the information one would wish to have, another parallel to everyday life.

In principle, case study tasks always lack details, so it is a part of the task to find out what the concrete case study is about. This implies that there is neither a right question nor a right answer. Instead, it is about the process of finding a solution, considering various questions and different competing approaches. There are no right or wrong solutions in case studies. What is more important is how the proposed solution is justified.

The purpose of analysis is not to reproduce learned knowledge, e.g., to explain a method or concept in detail, but rather to apply knowledge and experience to a decision-making situation, i.e., transfer. When working on the case study, it is important to adhere to academic standards. This includes the use of subject-specific literature beyond the coursebook, the ability to cite correctly, and the preparation of a bibliography.

Formal Guidelines and Submission Requirements

In addition to the information provided above, the case study has other formal components that must be integrated into every case study. It consists of the following parts (in this order):

- title page
- table of contents
- list of figures and/or tables (if necessary)
- list of abbreviations (if necessary)
- text part with introduction, body, conclusion
- bibliography
- list of appendices (if necessary)
- appendices and materials (if necessary)

Formal Document Requirements

The following formal requirements apply to the text of the case study.

Table 10: Formal IU Requirements for Case Studies

Length	7–10 pages of text
Paper size	DIN-A4
Margins	Top and bottom 2 cm; left 2 cm; right 2 cm
Font	General text: Arial 11 pt.; Headings: 12 pt., justified
Line spacing	1.5
Sentences	Justified, auto-hyphenation
Footnotes	Arial 10 pt., justified
Paragraph	6 pt. distance after line break
Section/subsection levels	Maximum three levels (1. Main heading, 1.1 Section, 1.1.1 Sub-headings)
	Only individual chapters in the text are numbered consecutively; otherwise, sections of the assignment, such as the list of figures or the bibliography, are not numbered.
	Do not use the underline function, and use italics sparingly to emphasize passages.

Source: Created on behalf of IU (2020).

The submission of the project report happens via the Turnitin portal. Instructions on how to submit your work can be found in a separate manual on myCampus. There you can also find out how the evaluation can be viewed directly on Turnitin after the publication of the grade on myCampus. It is not possible to deliver the information by email or by other means.

It is important that the affidavit first be submitted electronically via myCampus. Prior to this, it is not possible to submit your case study. Further information on this can be found in the "Instructions for Submission in myCampus—Turnitin".

Task

For the case study several topics are proposed. Details on the topic and task of the case study are provided in the respective courses. Any questions regarding the task can then be discussed with the tutor.

Students have the option to make use of any one of several opportunities to get support for their case study analysis with the course tutor. Taking advantage of these opportunities is the responsibility of the student and the use of these services is voluntary. It is possible to contact the tutor regarding formal and general questions about working on the case study. Please note: a review of outlines and aspects of the presentation is not intended here, since the student's ability to work independently is part of the evaluation and counts as a part of the overall assessment. There are however general tips for developing the case study to help you getting started.

Further specification and guidelines can be found in the respective Exam Guide in myCampus and the guidelines of particular courses.

UNIT 8

ACADEMIC WORK AT IU: THE BACHELOR THESIS

STUDY GOALS

On completion of this unit, you will have learned ...

- how a bachelor thesis at IU is structured.
- which instructions to use to write a bachelor thesis.
- which formalities need to be observed when writing a bachelor thesis.
- the requirements encompassing both the scope and content of a bachelor thesis.

8. ACADEMIC WORK AT IU: THE BACHELOR THESIS

Case Study

Simon can hardly believe it. Just six more months and his studies are done with the completion of his bachelor thesis. So many ups and downs, so many new experiences, so much effort given to balance the multiple demands of work, study, family, and leisure time. According to lecturers and professors, the bachelor thesis crowns the academic qualification process with an application-oriented question that is ideally relevant to one's own career and life. In the thesis, new findings are to be developed that apply to one’s vocation. Simon therefore chooses the following topic for his bachelor thesis: “The importance of digital word-of-mouth propaganda in marketing, using the example of the Daily Greens superfood brand”.

Simon chooses empirical methodological triangulation for this purpose. This means that he develops an online questionnaire which he will send via the Student Administration to a representative sample of students from his study program. In addition, Simon will conduct five qualitative, guided interviews with fellow students from his class who have Twitter, Facebook, and Instagram accounts. Simon will use both empirical methods to compile the findings and interpret them for his own company using this example of super foods.

8.1 The Bachelor Thesis at IU

The bachelor thesis represents the completion of one’s academic studies. Typically, the knowledge acquired during the studies is processed in a comprehensive academic work and new knowledge is generated on the basis of an investigation in a specific subject area. The bachelor thesis often poses great challenges for students, since the vast majority of the work is done independently, albeit under the supervision of the professor. In addition to a high degree of discipline and self-organization, conscientiousness also plays a special role when working on one’s bachelor thesis. This academic work not only represents the final step of the academic study program, but also offers students the opportunity to link theory with relevant questions of professional practice in such a way that the results can also be used professionally. For many students, the thesis is also an important contribution to career development.

It is therefore of great importance to familiarize yourself with the Thesis Handbook found in the myCampus course “Information Thesis in General”. It contains all essential information about the bachelor thesis. The handbook is a detailed orientation aid for the writing of the thesis for IU's distance learning program. With the bachelor thesis, students have the opportunity to identify a topic, develop a research question, and work out answers to the research question based on theory, rules, and theme-specific literature. Students must also demonstrate the ability to carry out research methodically convincing and be able to present the results of the study both in writing and orally.

The following reviews the process from the preparation of the bachelor thesis to its submission. More detailed information on finding a supervisor, the registration process, the corresponding processing deadlines, and the like can also be found in the Thesis Handbook.

General Learning Objectives

A bachelor thesis is a work of academic research that includes both theory and practical application. It includes identifying a topic that is relevant for a specific research area. The final thesis should show that students are able to independently work on a topic from one of the subject areas of the degree program within a given period of time, both in its technical details and interdisciplinary contexts, by using scientific methods. In addition, students must be able to present and defend the thesis in front of a commitee. A thesis can only be successfully completed if (1) there is real interest in the topic, (2) there is willingness and determination to become an expert in the subject area of the thesis, and (3) students fully support their respective research project. By working on the thesis, students will be able to identify, develop, carry out, analyze, and present a research project within the timeframe specified.

Structure

The following explanations are intended to give an overview of the structural elements of a bachelor thesis. Adhere to the following list in the order provided:

- front page
- lock flag (optional)
- acknowledgments (optional)
- abstract
- table of contents
- list of figures/tables (if required)
- list of abbreviations (if required)
- body
- bibliography
- appendix
- glossary (optional)
- affidavit (also called declaration of authenticity)

In rare cases, the supervisor can request a different thesis structure. In this case, it is necessary to coordinate accordingly.

The **title page** is the first page seen by the reader. Apart from the information listed below, nothing else should appear on the title page. The following should be included:

Title page
The title page is the first page of a bachelor's thesis and contains specific, mandatory information.

- full name of the university (no abbreviations, even if the logo is present)

- name of the degree program (no abbreviations)
- title of the thesis
- name of the author
- matriculation number
- address of the author
- name of the first supervisor
- submission date

Acknowledgement page
The acknowledgement page enables the author to thank all those who contributed to the success.

The **acknowledgement page** serves to acknowledge those who helped during the creation of the work. An acknowledgement page is not mandatory. However, if it becomes part of the work, the following should be observed. The bachelor thesis will be published and made publicly accessible over a long period of time. For this reason, attention should be paid to who is acknowledged. Usually students acknowledge their parents, their supervisor, data sources (e.g., persons interviewed), and the proofreaders. The thesis reflects the hard work from the students and various supporters and for this reason it should not be a list of friends. The acknowledgement page typically does not go beyond one page and is usually much shorter.

Abstract
An abstract allows the author to describe the essential content in a maximum of 200 words.

The **abstract** usually consists of a paragraph summarizing the main objectives, results, and conclusions. It should contain about 200 words and should not be longer than a page. Furthermore, it is recommended that keywords be written under the paragraph. Keywords are three to seven words that allow the reader to identify the topic.

The table of contents should list each chapter and each sub-chapter with the corresponding page numbers. If it is necessary to create a sub-section, at least two sub-chapters have to be added to the section.

Poor example:

1. Introductory remarks
   a) Structure of a distance learning course
2. Literature analysis

Better example:

1. Introductory remarks
   1.1 Definition of distance learning
   1.2 Forms of distance learning
   1.3 General rules of distance learning
2. Literature analysis

The level of detail for the table of contents should be discussed with the supervisor. As a rule, however, it is recommended not to have more than three levels of headings. Too many levels, such as 2.3.4.5.1, should be avoided. The chapters and sub-chapters must match the title and numbering of the text.

The list of tables/figures, abbreviations, and glossary are intended to help the reader find relevant additional information. If three or more tables are used, including a list of tables/figures is mandatory. Traditionally, the lists of tables and figures as well as the list of abbreviations are included at the beginning of the thesis, and the glossary is listed after the appendices.

The following section explains the main part of the thesis with the recommended points in more detail. The main part should contain the following topics, if not otherwise agreed with the supervisor:

- introduction (normally referred to as Chapter 1)
- literature review (usually referred to as Chapter 2)
- research methodology (usually referred to as Chapter 3)
- research findings (usually referred to as Chapter 4)
- conclusion (normally referred to as Chapter 5)
- recommendations/limitations (can be a separate chapter, but can also be included in Chapter 5)

The introduction should include a clear description of the purpose, aims and objectives of the thesis, which are expressed by means of research objectives and research questions. In addition, an overview of the overall structure of the thesis should be presented.

The **literature review** should in turn reflect a critical examination of literature relevant to the topic. Relevant literature includes relevant textbooks, reference books and articles from academic journals. Students can decide for themselves how to structure the literature analysis.

Literature review
Here, the current state of research on the research question of the thesis is discussed.

The thesis must include a chapter explaining the applied research methods (empirical research with qualitative or quantitative methods, literature and review work or mixed methods).

The research results should be presented in an individual chapter. It is advisable to use tables and visualisations here. The results are only presented here. At this point, no interpretation takes place. This is followed by the interpretation of the results. Here it is important that they are linked to the relevant literature mentioned in the theoretical foundation as well as to any empirical results of one's own. Furthermore, recommendations for action for the object of study are usually formulated in this chapter, which can be derived from the research process.

A conclusion (with an outlook) concludes the paper. Here, recommendations for further research and limitations that have arisen during the research process can be considered.

The **appendix** serves to present information that is too detailed for the body, but remains important for its understanding. These can be the original copies of the surveys, large tables, or scanned materials and transcripts of in-depth interviews. It has to be agreed with the respective supervisor whether interviews have to be transcribed and added. In general, questionnaires, transcripts, and other information can be attached in the original language unless the supervisor decides otherwise.

Appendix
This section contains all documents that are used for deeper understanding and closer examination.

Each appendix shall be designated with corresponding ordering, e.g., Appendix A, Appendix B. The pages in the appendix are numbered but not included in the 40 pages of the body. The body of the thesis should refer to each appendix.

Affidavit
This declaration is placed at the end of the Bachelor's thesis.

The **affidavit**/declaration of authenticity must be included in the thesis. It should be the last page of the thesis, inserted after the appendices. If the document is not included in the paper, the thesis cannot be evaluated. The declaration can be sent to the exams office by post within seven working days, otherwise the thesis will be graded as “failed”.

Formal Requirements for the Text

The following formal requirements apply to the text of the bachelor thesis:

Table 11: Formal IU Requirements for Bachelor Theses

Length	Dependent on course of study
Paper size	DIN-A4
Margins	Top and bottom 2 cm; left 2 cm; right 2 cm
Font	General text: Arial 11 pt.; Headings: 12 pt., justified
Line spacing	1.5
Sentence structure	Justified, auto-hyphenation
Footnotes	Arial 10 pt., justified
Paragraphs	6 pt. distance after line break
Chapter/section levels	Maximum three levels (1. Main heading, 1.1 Section, 1.1.1 Subheadings)
	Only individual chapters in the text are numbered consecutively; otherwise, sections of the assignment, such as the list of figures or the bibliography, are not numbered.
	Do not use the underline function, and use italics sparingly to emphasize passages.

Source: Created on behalf of IU (2020).

For the supervision of the bachelor thesis, the supervisor is generally available to give feedback and to answer questions that may arise during the writing of the thesis. It is, however, independent academic work and thus, is the responsibility of the student. The student should keep in mind that the thesis will be written for their distance learning program. This necessitates a high degree of self-organization and professionalism. Students must prepare for telephone appointments and be able to send in all materials in advance via email. As in modern professional life, support is provided by a virtual team via email and Teams/Zoom.

Further specification regarding the bachelor thesis as well as the Thesis Handbook can be found in the course “Information Thesis General” in myCampus.

UNIT 9

ACADEMIC WORK AT IU: ORAL ASSIGNMENTS

STUDY GOALS

On completion of this unit, you will have learned ...

- how to structure a successful oral assignment.
- the goal of using good supporting visuals.
- which principles lead to a visually appealing presentation.
- how verbal and non-verbal communication influence a presentation.
- about the evaluation criteria used to judge the success of a presentation.

9. ACADEMIC WORK AT IU: ORAL ASSIGNMENTS

Case Study

Maike has completed a training at the RheCom Academy where she focused on learning more about rhetoric and communication to help her with her terrible stage fright and reticence for public speaking. During her training she learned that presentations are as much about body language and other types of non-verbal communication as they are about the words being spoken. For the first few days at RheCom, they dealt with breathing techniques and the tone, speed, and variation of speech. Then they focused on practicing the delivery of a presentation, something Maike found to be incredibly helpful. She sees the value in learning these skills to be able to properly express herself in both professional and academic situations. Within the framework of the bachelor colloquium, it is important to be able to be convincing to a reader or audience while depicting scientifically-founded connections. She can prepare for the colloquium with special presentations during her studies.

9.1 Oral Assignments at IU

With an oral assignment, the objective is to present your work content with appropriate and appealing visuals in order to share methodological and technical know-how with the audience. With enough preparation and practice, you can confidently present your work. These skills are always in demand in one's professional life as often, you are required to prepare and present a concrete topic to a specific audience.

Objectives and Procedures

In this presentation, students demonstrate that they are able to work independently on an academic topic, and that they can showcase the research in an understandable and accessible way for the audience.

At the beginning of every presentation, as with any written assignments, the student should clearly state the research topic and relevant literature and, from this foundation, be able to present, and theoretically substantiate one's own work. Subsequently, the developed content must be presented within the given framework in a visually appealing way for the target audience. The presentation should be completed and presented in its entirety, following a straightforward outline, within the given time allowed.

Structure

The concrete structure of an oral assigment is determined by the topic and its parameters. However, all presentations have the same basic components in common:

- title slide
- outline
- introduction
- body
- conclusion
- list of figures/tables (if necessary)
- bibliography

Each content related part of the presentation has a different focus and goal, which should be taken into account when preparing (Polonsky & Waller, 2004). The introduction identifies the subject of the presentation, its limitations, and the outline of the presentation to follow. The introductory remarks should grab the audience's attention and jumpstart their interest; therefore, it is advisable to choose a concrete introduction to the subject in the form of a daring thesis, a picture, quote, question, or the like.

For the body of the presentation, the now-attentive audience should follow the interesting and sensibly-structured, comprehensible presentation. In order to achieve this goal, the contents must have a common thread woven throughout. The presentation of the topic of investigation follows the criteria of academic work in presentations, this means that the content presented verbally must also be traceable and verifiable, following with academic standards.

The conclusion of this presentation serves to draw one's own conclusions on the basis of an independent analysis of the subject matter and, if necessary, to provide new perspectives. Arguments are concluded with a synthesis of what's been said. Additional mandatory parts of the structure are the title slide, the outline and the bibliography at the end.

Supporting Visuals

Unlike classical speeches, presentations focus on facts and content. A good visualization of the topic is therefore essential (Polonsky & Waller, 2004, pp. 436–437; Seifert, 2015) because it should support the factual presentation. In the case of an oral assignment, supporting visuals can be created with the help of PowerPoint software.

To be convincing, visuals must complement and support the content being presented verbally. Therefore, the following four principles should be taken into account when creating each slide:

1. Visuals help foster understanding, but they are never an end in themselves! Animations are only appropriate if they increase understanding of the content being presented.
2. Messages should be used instead of headlines. This means that each slide should only deal with one core topic at a time and contain a well-considered "action title".

3. A verbal description should be given before an image is shown. Before moving to the next slide, presenters should tell their audience what comes next. This ensures that a common thread or theme throughout the PowerPoint presentation is maintained and that the audience is not subjected to abrupt changes in topic.
4. PowerPoint slides should be readable. The design of the slides, including the color, font type, and font size, must enable easy recognition of the visual material being presented. The slides should not be overloaded with too much information.

For students enrolled in bachelor degree programs, it is recommended that they refrain from experimenting with visuals if it is unclear how the audience is likely to respond. Instead, it is better to use pre-designed PowerPoint slide templates with suggested fonts and colors.

If the above principles are not observed, the presentation may not achieve its goals, as the audience may become distracted or, in the worst case, lose interest. For help in designing an appealing PowerPoint presentation, there are countless guidebooks and reference works available (Hüttmann, 2018; National Conference of State Legislatures, 2017) as well as many good and bad examples on the internet. Ultimately, however, the specifics of a presentation are a matter of personal taste.

Rhetoric and Appearance

For a presentation to be successful, verbal and non-verbal communication, including the speaker's posture, must convey an impression of credibility. It is not only what is said that matters, but how it is said.

With respect to verbal communication, the following four points are particularly important to being perceived as convincing:

1. Voice: The easiest way for us to speak is in our individual speech pattern. This is the vocal range we use when we speak effortlessly over a sustained period of time and which also varies. It conveys an authentic sound to the audience.
2. Pronunciation (articulation): The clarity of pronunciation, otherwise referred to as the sharpness of articulation, is also particularly important to comprehensibility.
3. Emphasis (accentuation): Good speakers speak in a varied manner. They consciously alter their voice by adjusting the volume, speed, and intonation to suit what is being said. The well-considered use of emphasis brings a presentation to life, helping to sustain audience interest.
4. Breathing: Good breathing is necessary to being able to effortlessly maintain speech. Additionally, a good breathing technique helps to maintain an adequate speaking speed, including adequate pauses for breath.

In addition to the verbal and non-verbal communication techniques already discussed, the following non-verbal signals contribute to a successful presentation:

1. Posture and appearance: The image conveyed by a speaker to an audience occurs through a speaker's posture, which is automatically projected onto the audience. If the speaker seems bored or tense, for example, this affect will be projected onto the

audience. Good speakers take advantage of this phenomenon by consciously paying attention to their posture, attempting to appear relaxed and confident. Paying attention to appearance includes choice of clothing, given that what we wear also influences how we appear.

2. Gestures: Hand gestures should accompany and support a presentation. Good speakers are aware of their hand gestures and use them in a targeted manner. In order that hand gestures appear natural, even when they are deliberately used, some practice on the part of the speaker is required. It is also helpful to observe the gestures of other speakers, noting their effect on the audience.
3. Facial expressions: These are facial gestures, so to speak. Facial expressions are individual, but they can also be used consciously to indicate and trigger emotions. Caution: Unconsciously, our facial expressions often show others what we are "really" thinking, and this can stand in stark contrast to what has been said.
4. Eye contact: Even in professional presentations recorded online, eye contact must be established between the speaker and the audience. This is best achieved when the camera is seen as an imaginary conversation partner and consciously looked at from time to time.

The communication techniques discussed here are individualized and their successful use requires good practice. If you want to check your posture, facial expressions, and gestures, it is best to stand in front of a large mirror. The presentation itself, with all its technical, visual, and rhetorical facets, should also be rehearsed several times. A mirror can also act as a training partner. However, more direct feedback can be obtained from another person.

Formal Requirements and Evaluation Criteria

Oral assignments are recorded using the Bongo tool. Information on how to register and use the tool are described in detail in the Bongo user manual.

An oral assignment lasts 15 minutes. During the presentation, the speaker has to take care to stay in time. A PowerPoint presentation, which must be converted into a PDF file before uploading to the Bongo tool, provides visual support to the presentation. Important: Unlike a colloquium, there is no discussion at the end.

It should be noted that in terms of content, the presentation should be complete, follow a clear structure, and abide by the given time limit. The core parts of the presentation are the introduction, the structure, the quality of argument and the conclusion, which when weighted together equal 70 percent of the grade. The rhetorical features and supporting visuals used are weighted at 30 percent, and are, therefore, important components of the oral assignment.

Further specification and guidelines can be found in the respective Exam Guide in myCampus and the guidelines of particular courses.

UNIT 10

ACADEMIC WORK AT IU: ORAL PROJECT REPORTS

STUDY GOALS

On completion of this unit, you will have learned ...

- how to structure a successful oral project report.
- why using graphics in an oral project report is important.
- which principles lead to a visually appealing oral project report.
- how verbal and non-verbal communication influence an oral project report.
- about the evaluation criteria used to judge the success of an oral project report.

10. ACADEMIC WORK AT IU: ORAL PROJECT REPORTS

Case Study

It is a big challenge for Simon to present the entire process of a project, and its results, to a large audience. It has therefore been particularly important for him to follow a clear structure throughout his projects in order to then present the results, findings, and future uses to a larger audience. It has also been important for Simon to show visual representations of his projects throughout various presentations. Throughout the course of many projects, he created photo logs so that he had some good, descriptive material that he could use during his oral project reports. Looking back, however, he is particularly proud of the fact that he chose to begin so many of his presentations with an interactive introduction, which turned out to be very successful. By creating a “reflective team” through a short, interactive audience survey, he not only had the full attention and focus of the audience, but it also helped quell his nerves, given that within the first few minutes of the presentation he lost the feeling of being directly at the center of attention.

10.1 Oral Project Reports at IU

Objectives and Procedures

The oral presentation of an independently designed, implemented, and documented project stands at the center of an oral project report. A successful presentation demonstrates that students are able to transfer their theoretical knowledge into practice within the framework of a specific project. Students also demonstrate that they can prepare and present a project in such a way that it becomes understandable and accessible to an academic audience. The main aim of a presentation is therefore, to familiarize the audience with the process followed in the project and to present the project’s results.

A topic must be identified and planning of the project must take place during the early stages of preparing a presentation, which is then followed by implementation of the project and its documentation. Once the overall project has been completed, individual aspects of the project are to be structured in such a way that they can be understood by an audience. Only at this stage are the actual presentation and supporting visuals created and prepared. Particular care must be taken to ensure that both the process followed in the project and the results are clearly formulated and presented.

Structure

In terms of content, the structure of a presentation is determined by how different phases of the project were carried out. However, all presentations have the same basic components in common:

- title slide
- outline
- introduction
- body
- conclusion
- list of figures/tables (if necessary)
- bibliography

Each content related component has a particular focus and a goal that should be taken into account when preparing the topic (Polonsky & Waller, 2004). The Introduction serves as a guide to the project, explaining the topic and briefly introducing the structure of the presentation. Importantly, the introduction should arouse attention and interest. Therefore, it is advisable to choose an engaging way to introduce the subject of the presentation, such as a daring thesis, image, or quote.

A description of how the project was implemented alongside its results should be given in the body of the presentation. Although each project is individual, a presentation can be structured based on the following questions:

- What was or were the goals of the project?
- How was the project completed? What preparatory measures were taken (data collection, data analysis, material evaluation, etc.)?
- What planning/organizational activities were necessary?
- Why was the project conducted in the way it was?
- Which methodology or other tools were chosen and why?
- Who were the stakeholders?
- Was a risk analysis undertaken? If so, what were the results?
- What were the main planning or project phases?
- How was the project structured?
- How did the project progress?
- How was the project scheduled? When did it start and end?
- What resources and costs were involved?
- Were the objectives achieved? What results were achieved? Did the results correspond with the goals of the project? If not, why not?
- What conclusions can be drawn from the project? What could be improved if necessary?
- If applicable, what reflections arose from the project with respect to deadlines and overall results?
- If applicable, what reflections arose from the project with respect to required resources and costs?

Note: The rules for academic work also apply to these presentations, meaning that the results presented in an oral presentation must also be verifiable and replicable.

The presentation should end with a conclusion that provides answers to all questions raised throughout the presentation and that reflects on the objectives achieved in the project. If it makes sense, a final review of potential follow-up projects can also be given. Additional mandatory parts of the structure are the title slide, the outline and the bibliography at the end.

Supporting Visuals

Oral project reports focus on the process followed during a project and the project's results. Visuals are presented with the help of software such as PowerPoint. Individual PowerPoint slides should illustrate the process of a project and the project's results in the truest sense of the word. It is therefore advisable to also use graphics or the SmartArt tool in PowerPoint in a targeted manner.

In order to be convincing, visuals must complement and support the content being presented verbally. Therefore, the following four principles should be taken into account when creating each slide:

- Visuals help foster understanding, but they are never an end in themselves! Animations are also only appropriate if they increase understanding of the content being presented.
- Messages should be used instead of headlines. This means that each slide should only deal with one core topic at a time and contain a well-considered "action title".
- A verbal description should be given before an image is shown. The audience must be verbally prepared by the presentation of slides before they are presented. This ensures that a common thread or theme throughout the PowerPoint presentation is maintained and that the audience is not subjected to abrupt changes in topic.
- PowerPoint slides should be easily readable. The design of the slides, including the color, font, and font size, must enable easy recognition of the visual material being presented. PowerPoint slides should also not be overloaded with too much information.

For students enrolled in bachelor degree programs, it is recommended that they refrain from experimenting with visuals if it is unclear how the audience will respond. Instead, it is better to use pre-designed PowerPoint slide templates with suggested fonts and colors.

If the above principles are not observed, the presentation may not achieve its goals, as the audience may become distracted or, in the worst case, lose interest in the topic being presented. For help in designing an appealing PowerPoint presentation, there are countless guidebooks and reference works available (e.g., National Conference of State Legislatures, 2017) as well as many good and bad examples on the internet. Ultimately, however, the specifics of a presentation are a matter of personal taste.

Rhetoric and Appearance

For a presentation to be successful, verbal and non-verbal communication, including the speaker's posture, must convey an impression of credibility. It is not only what is said that matters, but also how it is said.

With respect to verbal communication, the following four points are particularly important to being perceived as convincing:

1. Voice: The easiest way for us to speak is in our individual speech pattern. This is the vocal range we use when we speak effortlessly over a sustained period of time and which also varies. It conveys an authentic sound to the audience.
2. Pronunciation (articulation): The clarity of pronunciation, otherwise referred to as the sharpness of articulation, is also particularly important to comprehensibility.
3. Emphasis (accentuation): Good speakers speak in a varied manner. They consciously alter their voice by adjusting the volume, speed, and intonation to suit what is being said. The well-considered use of emphasis brings a presentation to life, helping to maintain audience interest.
4. Breathing: Good breathing is necessary to being able to effortlessly maintain speech. Additionally, a good breathing technique helps maintain an adequate speaking speed, including adequate pauses for breath.

In addition to the verbal and non-verbal communication techniques already discussed, the following non-verbal signals contribute to a successful presentation:

1. Posture and appearance: The image conveyed by a speaker to an audience occurs through a speaker's posture, which is automatically projected onto the audience. If the speaker seems bored or tense, for example, this affect will be projected onto the audience. Good speakers take advantage of this phenomenon by consciously paying attention to their posture, attempting to appear relaxed and confident. Paying attention to appearance includes choice of clothing, given that what we wear also influences how we appear.
2. Gestures: Hand gestures should accompany and support a presentation. Good speakers are aware of their hand gestures and use them in a targeted manner. In order that hand gestures appear natural, even when they are deliberately used, some practice on the part of the speaker is required. It is also helpful to observe the gestures of other speakers, noting their effect on the audience.
3. Facial expressions: These are facial gestures, so to speak. Facial expressions are individual, but they can also be used consciously to indicate and trigger emotions. Caution: Unconsciously, our facial expressions often show others what we are "actually" thinking, and this can stand in strong contrast to what has been said.
4. Eye contact: Even in professional presentations recorded online, eye contact must be established between the speaker and the audience. This is best achieved when the camera is seen as an imaginary conversation partner and consciously looked at from time to time.

The communication techniques discussed here are individualized and their successful use requires good practice. If you want to check your posture, facial expressions, and gestures, it is best to stand in front of a large mirror. The presentation itself, with all its technical, visual, and rhetorical facets, should also be rehearsed several times. A mirror can also act as a training partner. However, more direct feedback can be obtained from another person.

Formal Requirements and Evaluation Criteria

Oral project reports are recorded using the Bongo tool. Information on how to register and use the tool are described in detail in the Bongo user manual.

These presentations lasts 15 minutes. During the presentation, care must be taken to ensure that this timeframe is adhered to. A PowerPoint presentation, which must be converted into a PDF file before uploading to the Bongo tool, provides visual support to the presentation. Important: Unlike a colloquium, there is no discussion at the end of this presentation.

It should be noted that in terms of content, the presentation should be complete, follow a clear structure, and abide by the given time limit. The core parts of the presentation are the definition of the task, its structure, and the implementation of the project process, which when weighted together equal 35 percent of the grade. The originality of the approach taken to the solution is weighted at 20 percent and the quality of the solution(s) at 15 percent, which are both taken into account in the evaluation. The rhetorical features and supporting visuals used are weighted at 30 percent, and are important components of this presentation.

Further specification and guidelines can be found in the respective Exam Guide in myCampus and the guidelines of particular courses.

UNIT 11

ACADEMIC WORK AT IU: THE COLLOQUIUM

STUDY GOALS

On completion of this unit, you will have learned ...

- the components of a colloquium.
- the objective of a colloquium.
- what interdisciplinary evaluation criteria are applied in a colloquium.
- the tips and hints available to prepare yourself for this unique exam.

11. ACADEMIC WORK AT IU: THE COLLOQUIUM

Case Study

Simon and Maike have received initial positive feedback from their bachelor thesis supervisors about how their bachelor theses are coming along. Now they need to prepare for the colloquium. The colloquium summarizes the entire process and the essential contents and findings of the bachelor thesis so that students can explain and defend the knowledge they gained as well as show improved scientific performance. Simon and Maike are well aware that it is not only a question of presenting the process and the results, but also of preparing themselves for overarching questions and proposals for discussion on the part of the examiners. The examiners would like to see to what extent both students are able to connect logical and causal arguments for their work, as well as to discuss them controversially and place them within a larger framework. Therefore, both students use one another to practice the colloquium several times in advance. This is the perfect opportunity to time their presentation to make sure they do not go over their allotted timeframe. Both Simon and Maike have a red card to discreetly signal three to five minutes before the presentation time runs out, so that the speaker knows to finish as quickly as possible.

11.1 The Colloquium at IU

The colloquium occurs at the end of the bachelor program and thus marks the end of several years of intensive study on various subjects. It is of course a performance review, however its characteristics differ significantly from the oral exams taken during the course of study. This unit considers the unique position of the colloquium.

Objectives and Procedure

The colloquium takes place after the submission and positive assessment of the bachelor thesis and serves primarily as proof that the written thesis has been completed in full. There are two major components for this: (1) the presentation of the written work and (2) a discussion of the contents with the examiners.

The invitation to the colloquium is issued by the examiner of the bachelor thesis and is a sure sign that the bachelor thesis has been passed. Admission to this partial exam is granted by the Examination Office and the guidelines and forms deposited in myCampus should be read in advance of the bachelor thesis. The colloquium will be held either at a study site of IU, or online, independent of location, for example via Teams. In any case, it is important to consult the supervisor in a reasonable amount of time.

After the approximately 30-minute colloquium, the student is asked to leave the (virtual) room briefly so that the examiners can discuss and agree upon their evaluation. The student is then invited back in and receives direct feedback. After the successful completion of all modules, including the bachelor thesis and colloquium, the diploma is issued and sent by mail.

Presentation

The colloquium usually starts with a presentation. The presentation should show the whole research process, starting with the research question and finishing with the conclusion. This should usually last 15 minutes. To avoid problems, use Microsoft Office PowerPoint for the presentation.

Follow the typical three-part structure of a presentation so that the first part of the colloquium is roughly structured as follows:

1. Introduction
2. Body
3. Conclusion

After capturing the audience's interest through the introduction, one moves into the body of the thesis presentation. As this presentation is geared toward an academic audience, the content should be both intellectually demanding but also plausible and understandable. It is important to emphasize the innovative points of the written explanation and to only briefly refer to the starting point of the work, as the focus should be on one's own efforts and learning. Keep in mind that the entire research process and results should be presented; simply selecting certain aspects to share is not sufficient. The conclusion of the colloquium presents the findings of the bachelor thesis and, if possible, provides new perspectives. There should also be a synthesis of what has been said during the presentation, summarizing the student's argumentation.

Before the colloquium, it is advisable to discuss with the supervisor whether (subject-specific) particulars should be taken into account for the presentation.

Discussion

In the subsequent question-and-answer session, the aim is to defend both the bachelor thesis and the presentation. Since the primary purpose of a defense is to provide adequate linguistic and content-related justification for what has been said or written and for the conclusion reached, the most probable "why" questions are to be expected regarding the general topic of the thesis and the methodological approach.

Evaluation Criteria

Both components of the colloquium will be assessed on an interdisciplinary basis using the following four criteria:

1. Understanding and applying scientific analytical methods,
2. Structure and content of the presentation,
3. Ability to defend the bachelor thesis academically, and
4. Content and logical consistency of answers to exam questions.

From this it becomes clear that, in addition to pure subject-specific knowledge, methodological and rhetorical competences are also assessed. These abilities and skills develop continuously during a course of study and should be consciously trained and improved upon starting at the beginning of a student's time at university.

Notes for Preparation

The colloquium is a special performance review that students can start working toward from the very beginning of their course of study. The preparation therefore also differs significantly from other types of exams. Here are hints that will help you prepare.

1. Determine (and implement) your own preferences for supporting visuals. Although slide and presentation designs should follow certain basic rules, their concrete design can be individual and thematic, as well as subject-specific. In preparation for the colloquium, reflect on which aspects and design criteria are important to you and then—in compliance with general requirements—implement your personal preferences. This lends authenticity to the presentation and takes the individual learning processes into account.
2. Keep in the forefront of your mind your own individual work. It is important to have your own written bachelor thesis present. It is best to read the thesis again shortly before the colloquium and refresh any forgotten background knowledge.
3. Prepare discussions. The possible questions in the colloquium are probably the most feared part of this exam, both in terms of their content and the (virtual) face-to-face situation. It can be helpful while reviewing your thesis to think about likely questions and formulate answers. In addition, looking at the exposé of the bachelor thesis can also help: What did I want back then? What have I ultimately been able to achieve?
4. Have "standard work" at the ready. The written bachelor thesis is the basis of the colloquium, the "standard work" so to speak. In consultation with the supervisor, you should therefore have this printed and within reach during the colloquium.
5. Use technical jargon. During your studies, you read countless sources and listened to many experts. These experiences can prove highly useful here. It is advisable to review topic-specific terms, keywords, and name experts in the field, etc. in the run-up to the colloquium so that you can respond adequately to the discussion.
6. Consult with your supervisor. Colloquia are a little bit different depending on the supervisor, so you should always consult him/her to avoid possible misunderstandings. It is advisable to collect and discuss the questions on the individual sections of the bachelor thesis module. (Don't forget: document answers!)
7. Practice, practice, practice. The presentation part can (and should) be practiced as often as possible so that nothing stands in the way of a clear presentation.

Further information regarding the colloquium can be found in the Thesis Handbook.

UNIT 12

ACADEMIC WORK AT IU: PORTFOLIOS

STUDY GOALS

On completion of this unit, you will have learned …

- the types and varieties of portfolios.
- the advantages of an online portfolio.
- what goes into the portfolio evaluation.
- what role each phase of portfolio development has on the evaluation.
- how the formalities for the end product are defined.

12. ACADEMIC WORK AT IU: PORTFOLIOS

Case Study

Simon accomplished his goal and successfully completed his bachelor degree in business studies at IU. He is now an economist and happy that he has a few months to relax and just focus on work; however, he already knows that he would like to enroll relatively seamlessly into the Master's program in Marketing Management. His boss has bigger things planned for him and predicts that Simon will have a very active role in the future reorganization of the management of zielNET. Simon chose this Master's program because he often deals with marketing as well as management in his new role at work. Simon is pleased to discover that it includes a number of creative modules such as Digital Design and Design Thinking, as this is precisely where Simon would like to develop his skills further and gain actual experience and practice in operative design and in the creation of brands. In his annual review with his boss, in which they also discuss his new academic venture, his boss encourages him to start building a portfolio from the very beginning of his new program. The creative design processes require a lot of time, space, effort, and patience. Simon's boss explains to him how important it is to document his learning and development process in a digital format, i.e., in a portfolio. This portfolio can be used as a starting point in conversations with real design professionals. Critical in this phase is getting to know oneself as a designer, developing an individual style, and creating authentic, practical models and samples to showcase.

This advice sets Simon up to be able to answer the following questions when developing his portfolio:

- How can I compete and stand out from the crowd?
- How can I learn the necessary practical skills and study methods that are expected of me as a creative designer?
- How do I build up my personal brand and how does successful self-marketing work?
- How can I optimally present my acquired know-how and projects to future employers and clients?

12.1 Portfolios at IU

A preliminary note: A portfolio evaluation occurs in many subject areas. In this unit, the example used throughout this unit comes from the Bachelor in Communication Design program. This example was chosen because it provides the general requirements for the evaluation in such a way as to be transferrable to other areas of study.

Types and Sizes

(Self-)marketing in the context of the portfolio:

The portfolio is the most important self-marketing tool for designers, however it is also becoming increasingly relevant in other academic areas. A portfolio should be created and added to over the course of study. Relevant professional websites and journals provide a helpful overview of what should be considered when creating a portfolio and what advantages it can bring.

The portfolio, which primarily serves the purpose of advertising and "self-marketing", corresponds to the definition of a presentation portfolio according to Baumgartner (2012, pp. 49–54). Some advantages of an online portfolio compared to a hard-copy portfolio are outlined here:

- less effort and more time for the actual exciting projects,
- automatic content for personal marketing via social and professional networks,
- no knowledge of coding is needed with a good and intuitive tool,
- in contrast to offline portfolios, the online portfolio can be quickly forwarded to interested parties, potential employers, and clients, and
- it is the best possible way of showing potential clients your personality with little effort, allowing you to stand out from the crowd.

There are further advantages to creating a portfolio within the specific evaluation framework provided by the university. In design studies, for example, students gain more practice conveying design through the use of practical concepts, methods, or tools. Courses of study with a strong connection to design are dependent on this relevant association and supportive reflection during the design processes. The individual learning and evaluation process should be documented and supported in ways that include reflection on the process itself. This type of portfolio corresponds to a reflection portfolio (Baumgartner, 2012, pp. 49–54).

With the help of a portfolio, students can document their personal learning path through the submission of a diversity of assignments that constitute their portfolios. As part of an online portfolio, students can present selected assignments and projects—from the initial idea to the finished product—to interested parties and potential clients. Similar to a model's photo portfolio, the students are accompanied through the development of their design projects by the lecturers, in an advisory capacity, from the first sketch to the finished product.

Documentation over time of a student's own development of qualifications and competencies in terms of their career is referred to as a development portfolio (Baumgartner, 2012, pp. 49–54).

In summary, the portfolios differ in their use and objectives, depending on their nature and structure:

- Presentation portfolios are primarily used for job applications and "self-marketing".
- Reflection portfolios document and reflect the individual learning and evaluation process.
- Development portfolios show the long-term development of qualifications and competencies with regard to one's professional career.

Evaluation Process

How is the evaluation of a portfolio at IU structured? In order to develop a product as part of the portfolio evaluation, a multi-stage iterative process is necessary. This is reflected in different phases and iterative processes. The basic structure of this evaluation usually follows the following pattern:

1. Conception phase: conceptualize/sketch/scribble/raw design
2. Development/reflection phase: definition of (design) parameters/digital design/intermediate step(s)/correction phase(s)/reflection
3. Finalization phase: End product

Along these phases, a total of three portfolio parts need to be submitted in this specific sequence and in the detailed format described in the guidelines.

The conception phase defines an initial concept with sketches, scribbles, or raw designs of the desired product. If possible, this is to be done with pen and paper as part of an initial brainstorming session. These sketches should be submitted to the tutor together with an initial concept in writing, no longer than half of a page (DIN A4).

After the tutor evaluates the sketches and concept and gives feedback for changes and improvements, the second phase of development begins. In this second phase, the parameters are defined and a first digital design is created with the appropriate software. In the communication design example, parameters would include aspects such as form, color, and font. Information on the desired parameters can be found in the corresponding guide of the respective course/module on myCampus. In the second phase, intermediate steps leading to the end product are developed. Depending on scope and complexity, several intermediate steps may be necessary and can, for example, continue to be accompanied in a reflective manner by online tutorials and support from the tutor.

The end product is created in the finalization phase. It is submitted together with a two-page abstract. The basis for this abstract is the original concept. Further formalities for the evaluation can be found in the guidelines.

Table 12: Overview of the Audit Performance Portfolio

Stage	Intermediate result	Performance to be submitted
Conception phase	Portfolio part 1	Concept presentation in text form (approx. 1/2 page); sketch / moodboard / draft, etc.
Feedback		
Development / reflection phase	Portfolio part 2	Explanation of implementation in text form (approx. 1/2 page), first digital draft / milestone / intermediate step
	Feedback	

Stage	Intermediate result	Performance to be submitted
Finalization phase	Portfolio part 3	Two-page abstract (making of), final product (action), digital signature (optional), linking OneDrive Business folder (incl. all files)
	Feedback + Grade	

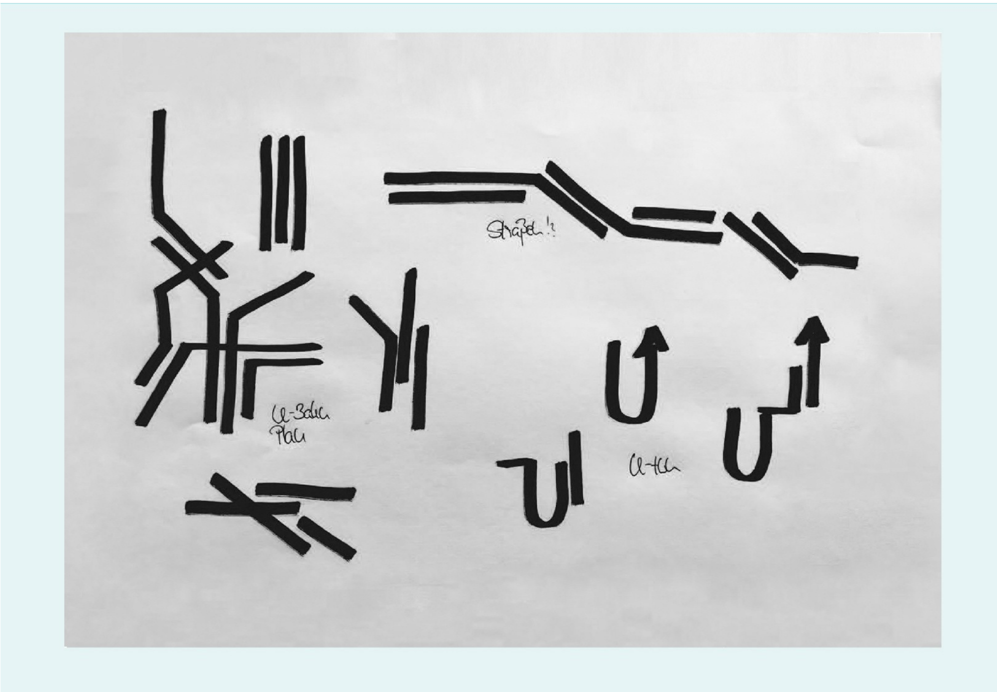
Source: Created on behalf of IU (2020).

Conception Phase

A new corporate design for the bicycle brand “URBAN DYNAMIC SYSTEMS” is being developed. Initial steps include online research to find inspiration that will shape the development process. For example, it is important to analyze topics such as the history of the company, relevant colors, shapes, and societal influences.

In addition, milestones of the design process are illustrated within the framework of the previously mentioned processes using the example of a logo design for URBAN DYNAMIC SYSTEMS. After the research, the creation of several first sketches (scribbles) should take place; see the figure below as an example.

Figure 10: Phases of the Design Process: Scribbles/Sketches and Logo



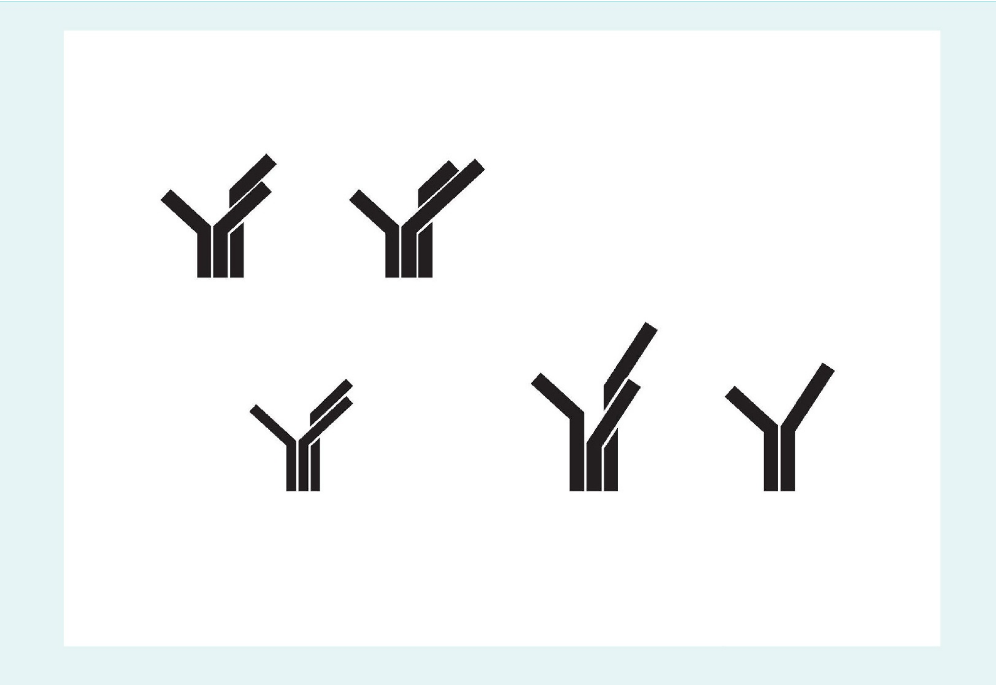
Source: Created on behalf of IU (2020).

These first sketches are supported by the design concept. After an evaluation and feedback by the tutor on how to proceed, the second phase follows.

Development/Reflection Phase

Now it’s time to create the first digital sketch using the recommended software.

Figure 11: Phases of the Design Process: First Digital Draft, Variations of Logo

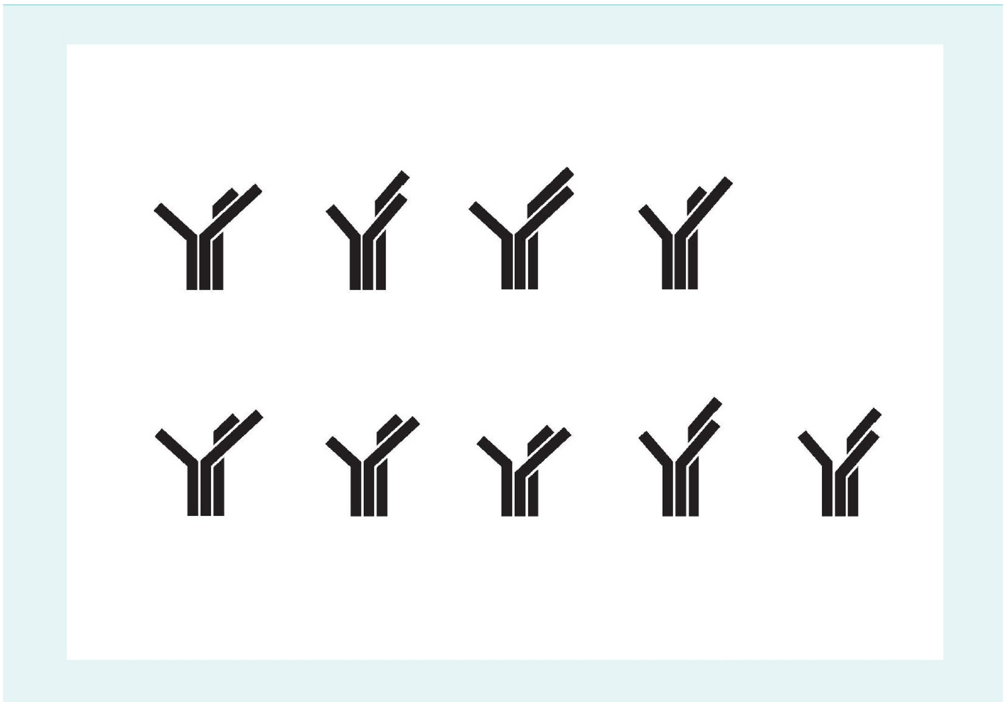


Source: Created on behalf of IU (2020).

After a short evaluation and feedback by the tutor on how to proceed, the development of the intermediate steps leading up to the completion of the final product in the finalization phase usually follows.

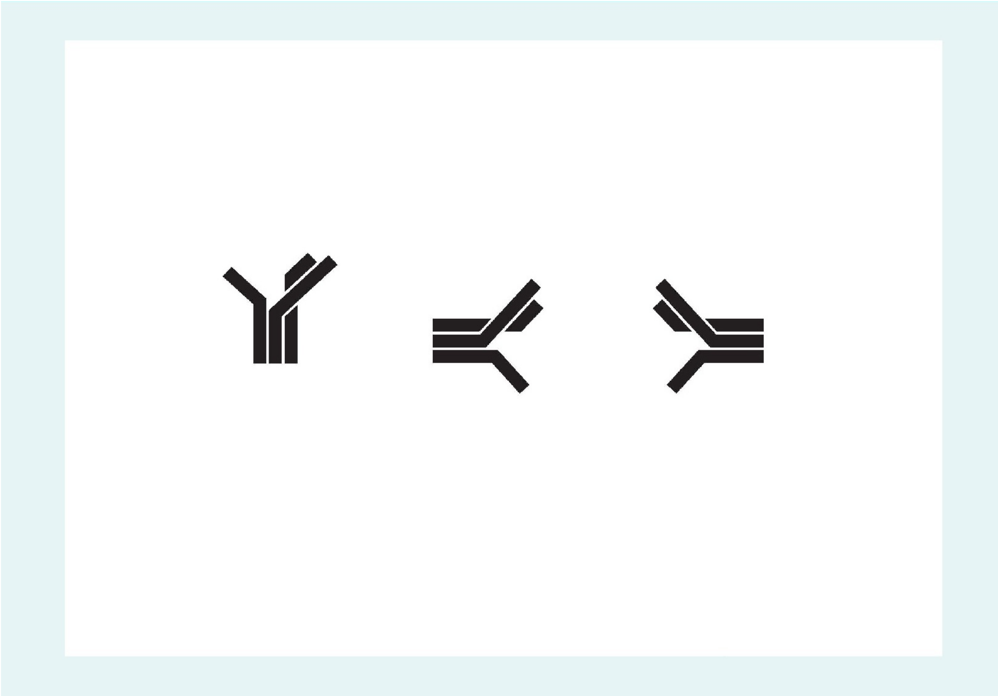
Further intermediate steps follow, as illustrated in the following figures.

Figure 12: Phases of the Design Process: Second Digital Draft, Detailed Variations of Logo



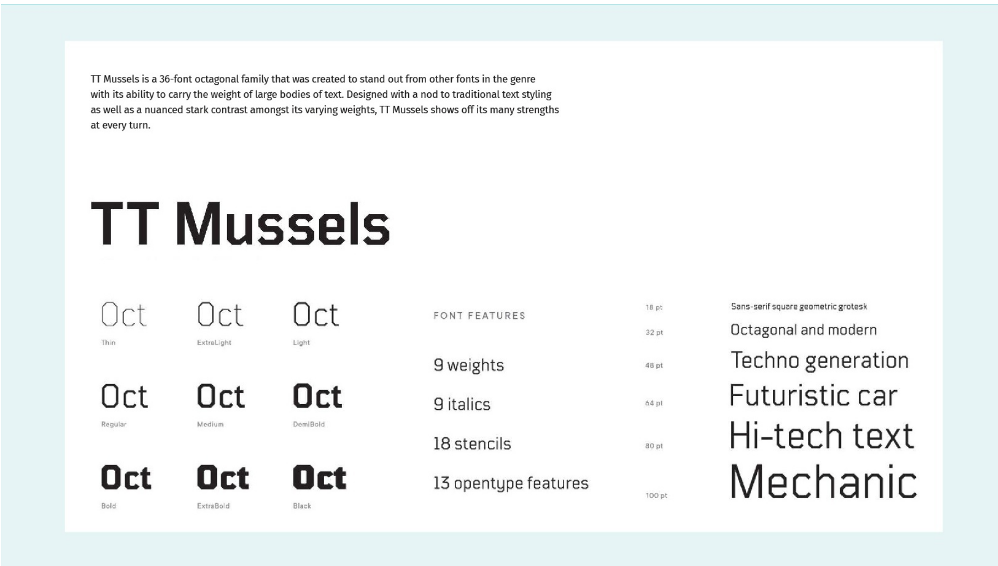
Source: Created on behalf of IU (2020).

Figure 13: Phases of the Design Process: Third Digital Draft, Rotated Variations of Logo



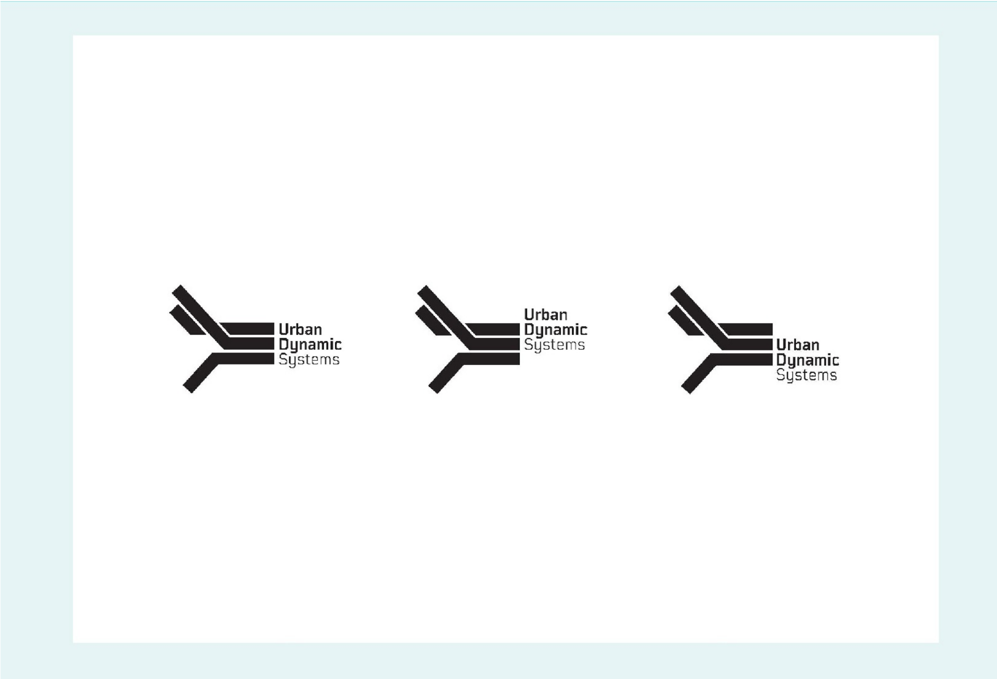
Source: Created on behalf of IU (2020).

Figure 14: Phases of the Design Process: The House Font



Source: Created on behalf of IU (2020).

Figure 15: Phases of the Design Process: Fourth Digital Draft, Variations of Logo with Text



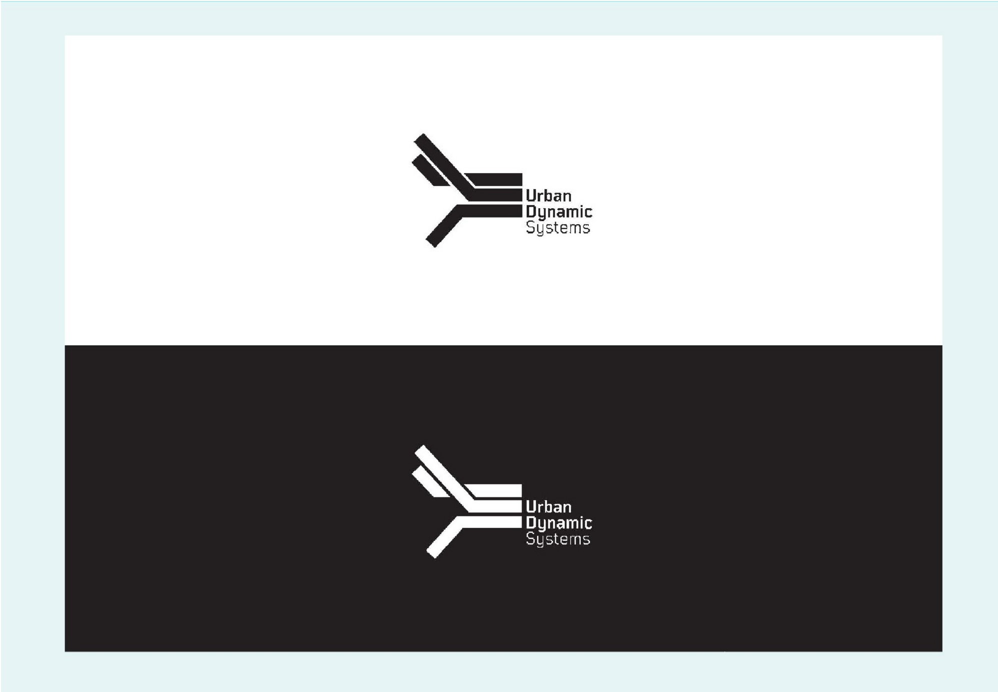
Source: Created on behalf of IU (2020).

Finalization Phase: The Final Product

The last milestone is the final product. The final product is submitted together with a two-page abstract. The basis for this abstract is the original concept. Furthermore, the final version should be prepared and supported with more information about the design process, the parameters, and the software used.

The following is an example of an overview of the selected typography and the final logo for the end product:

Figure 16: Phases of the Design Process: End-Product, Logo with Text



Source: Bilz & Schneider, 2017.

Formal Guidelines and Submission Requirements

The assignments are delivered as part of the portfolio evaluation via the PebblePad portal. Detailed instructions on how to submit work can be found in a separate manual on myCampus. It is not possible to submit the information by email or by other means.

Here, too, the affidavit has to be assented to.

Formal Requirements for the Abstract

The following formal guidelines apply to the text of the abstract:

Table 13: Formal IU Requirements for Portfolio Abstracts

Length	2 pages of text
Paper size	DIN-A4
Margins	Top and bottom 2 cm; left 2 cm; right 2 cm
Font	General text: Arial 11 pt.; Headings: 12 pt., justified
Line spacing	1.5
Sentence structure	Justified, with auto-hyphenation

Footnotes	Arial 10 pt., justified
Paragraph	6 pt. Distance after line break

Source: Created on behalf of IU (2020).

Task

One or more topics for the portfolio are proposed in the specific course. Typically, students are expected to complete a theory-based, conceptually-structured, and practice-oriented implementation of the product, ideally linked to the student's profession. Details on the topic and tasks of the portfolios are provided in the students' respective courses. Any questions regarding the assignment can then be coordinated with the respective tutor.

Generally, there are several channels open to a supervision of the portfolio. The respective use is in the own area of responsibility. The independent development of a product and the filling of the particular portfolio parts is part of the examination to be performed and is included in the overall assessment.

Further specification and guidelines can be found in the Exam Guide in myCampus and the guidelines of particular courses.

UNIT 13

ACADEMIC WORK AT IU: EXAMS

STUDY GOALS

On completion of this unit, you will have learned …

- out of all the different performance reviews, what is meant by the term “exam”.
- the difference between a module and a course.
- how an exam is structured.
- what type of exam you can take.

13. ACADEMIC WORK AT IU: EXAMS

Case Study

IU students, Meike and Simon, are approaching their final exams for their respective courses. They wonder how things will proceed. They find out that it is possible to either take the exam online at home or at an examination center on a specified date and time. Meike prefers the flexible option; she will complete the exam online—she has no problem with virtual testing and video interface and wants to do the exam once she is ready. For Simon, the idea of sitting in front of a camera and having to concentrate on the exam questions is pretty unsettling. He appreciates the fact that he can set up the exam dates himself at an actual physical location which will help to provide structure to his studying more easily.

13.1 Exams at IU

In this unit, an important form of examination at IU is explained in more detail—the exam as a performance review. The exam is one of the most central and common forms of testing knowledge at IU. It can be achieved with maximum flexibility, according to one's own preferences, and can be practiced in advance, thus reducing exam anxiety and deadline pressure (Fernández-Castillo & Caurcel, 2015). A written exam is used to test the students' knowledge of relevant subject matter and various questions are asked to test the students' knowledge acquisition.

Definition

A written examination "is to be carried out under supervision according to a set amount of time, whereby its scope and content as well as the degree of independent scientific achievement are determined by the course of study" (Theisen as cited in Koeder & Hamm, 1999, p. 255, author translation). Here, not only factual knowledge is delivered, but also "certain facts are to be processed in an application- and problem-oriented way" (Koeder & Hamm, 1999, p. 255, author translation).

General Information

There are certain guidelines for a written exam. The corresponding examiner determines and communicates the support materials allowed during the test. These, as well as other guidelines, such as approved legal texts, are set in myCampus (please consult the latest version). The entire content of the course book is covered in the exams. This serves as proof that the content of the course book has been learned. In addition, a transfer task is also required. This means that the knowledge learned from the course book needs to be applied independently and transferred to different cases.

The exam duration depends on the number of credits (ECTS) earned for the module/course:

- 5 ECTS in the module = 90 minutes exam duration
- 10 ECTS in the module = 2 examinations of 90 minutes each

Exceptions:

- 10 ECTS language courses (old): 180 minutes exam duration.
- 10 ECTS in the module: These modules in Master's programs usually consist of a written exam of and other exams.

The module manual provides an overview of the courses and the associated ECTS.

If an exam is the chosen type of performance review for a module, a sample exam, including a sample solution, is provided. If there is currently no sample in a module, it is in progress. A sample exam of this kind can be found in myCampus in the respective course and is for practical purposes only. An overview of the structure, format, as well as the point distribution used, is provided. However, these do not have the same extent of questions as actual exams. Moreover, the availability of such a model exam is not a prerequisite for passing the actual exam.

In the exam the respective task is always to be read very carefully! In single-choice questions, only one possible answer is correct. If several answers are ticked, no points are awarded. Both positive and negative questions can occur. In the case of open-ended questions, particular attention should be paid to single words and terms in the question. With the help of key words in the exam question, studens can detect the level of academic performance expected for the task at hand, examples are: name, describe, explain and illustrate.

How Can I Take an Exam?

The General Examination Regulation (*Allgemeinen PrüfungsordnungAPO*) set up in myCampus, stipulate that there are two types of examinations:

1. Written exam. Here, the exam must be written by hand on site at an examination or study center. The choice of the examination or study center is left up to the students. The same applies to the examination date, which is offered by IU twice a month. Actual dates and examination centers are listed in myCampus.
2. Online exam. Here, the exam is conducted online using live supervision (proctoring). The exam location and time can be determined individually by the students, however, this must comply with the guidelines of the online exam, found in myCampus.

Furthermore, there is the possibility of taking a free online trial exam with a supervisor (proctor). This does not include any subject-specific technical questions, but offers the opportunity to test the technical equipment and get more comfortable with the entire setting. IU strongly recommends taking advantage of this offer in order to avoid possible technical problems during the actual exam and to get to know the procedure of an online exam.

A trial exam without a proctor can be practiced at any time.

The questions for the written and online exams come from the same pool of questions. This means that any question from a written exam can also occur in an online exams—and vice versa.

Further Information

Further information about exams can be found in myCampus under the section “FAQ – Exams”.

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